

project_final

2025-05-18

Set Up

```
# This table contains the result of the interview, the household  
# factor and other household characteristics.  
household_data <- read.csv(unz("DATABASE.zip", "CS_HOGARES.csv"))  
  
# This table contains the socio-demographic characteristics of the  
# household members, health status and utilization of health services, among others.  
individual_data <- read.csv(unz("DATABASE.zip", "CS_RESIDENTES.csv"))  
  
# The table includes information on variables related to the health  
# status of adults, such as obesity, hypertension, depressive symptoms,  
# diabetes mellitus, reproductive health, among others, which were  
# collected in the health questionnaire for adults (20 years and older).  
adult_diseases_data <- read.csv(unz("DATABASE.zip", "CS_ADULTOS.csv"))  
  
# The table includes variables from section IV of the household  
# questionnaire and those corresponding to the questionnaire of health  
# service users, these variables capture situations regarding the use of  
# ambulatory services, accessibility and quality of health services,  
# medicines, among others.  
health_quality_data <- read.csv(unz("DATABASE.zip", "CS_SERV_SALUD.csv"))
```

Descriptives

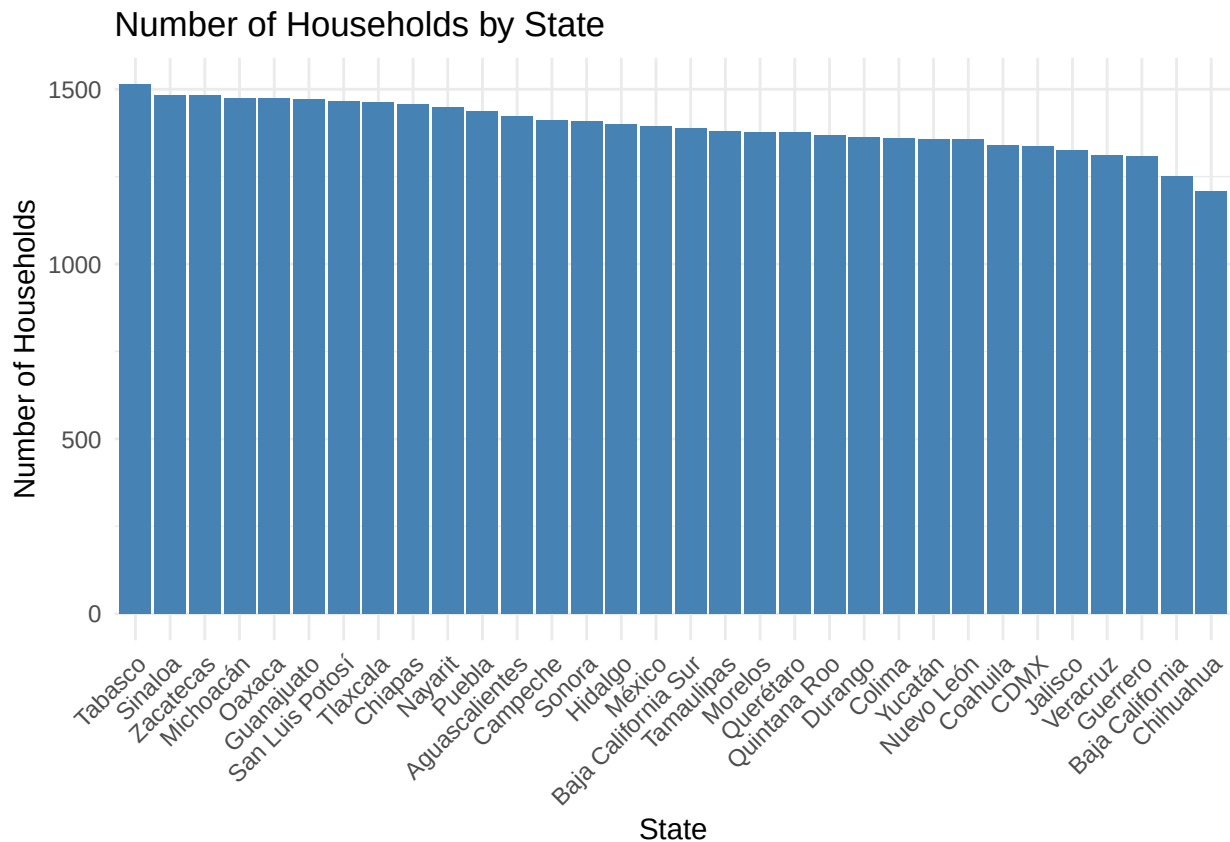
```
state_names <- c(  
  "1" = "Aguascalientes", "2" = "Baja California", "3" = "Baja California Sur",  
  "4" = "Campeche", "5" = "Coahuila", "6" = "Colima", "7" = "Chiapas",  
  "8" = "Chihuahua", "9" = "CDMX", "10" = "Durango", "11" = "Guanajuato",  
  "12" = "Guerrero", "13" = "Hidalgo", "14" = "Jalisco", "15" = "México",  
  "16" = "Michoacán", "17" = "Morelos", "18" = "Nayarit", "19" = "Nuevo León",  
  "20" = "Oaxaca", "21" = "Puebla", "22" = "Querétaro", "23" = "Quintana Roo",  
  "24" = "San Luis Potosí", "25" = "Sinaloa", "26" = "Sonora", "27" = "Tabasco",  
  "28" = "Tamaulipas", "29" = "Tlaxcala", "30" = "Veracruz", "31" = "Yucatán",  
  "32" = "Zacatecas"  
)  
  
# Count households by state and exclude NA values  
household_counts <- household_data %>%
```

```

filter(!is.na(ENT)) %>%
count(ENT, name = "num_households") %>%
mutate(state_name = state_names[as.character(ENT)])

# Plot
ggplot(household_counts, aes(x = reorder(state_name, -num_households), y = num_households)) +
  geom_bar(stat = "identity", fill = "steelblue") +
  labs(title = "Number of Households by State",
       x = "State",
       y = "Number of Households") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))

```



UPDATING THE DATABASES WITHOUT TOP 5 STATES

```

household_data <- household_data %>%
  filter(ENT != 2 & ENT != 9 & ENT != 14 & ENT != 19 & ENT != 22)

individual_data <- individual_data %>%
  filter(ENT != 2 & ENT != 9 & ENT != 14 & ENT != 19 & ENT != 22)

adult_diseases_data <- adult_diseases_data %>%
  filter(ENT != 2 & ENT != 9 & ENT != 14 & ENT != 19 & ENT != 22)

```

```
health_quality_data <- health_quality_data %>%
  filter(ENT != 2 & ENT != 9 & ENT != 14 & ENT != 19 & ENT != 22)
```

DESCRIPTIVE STATISTICS

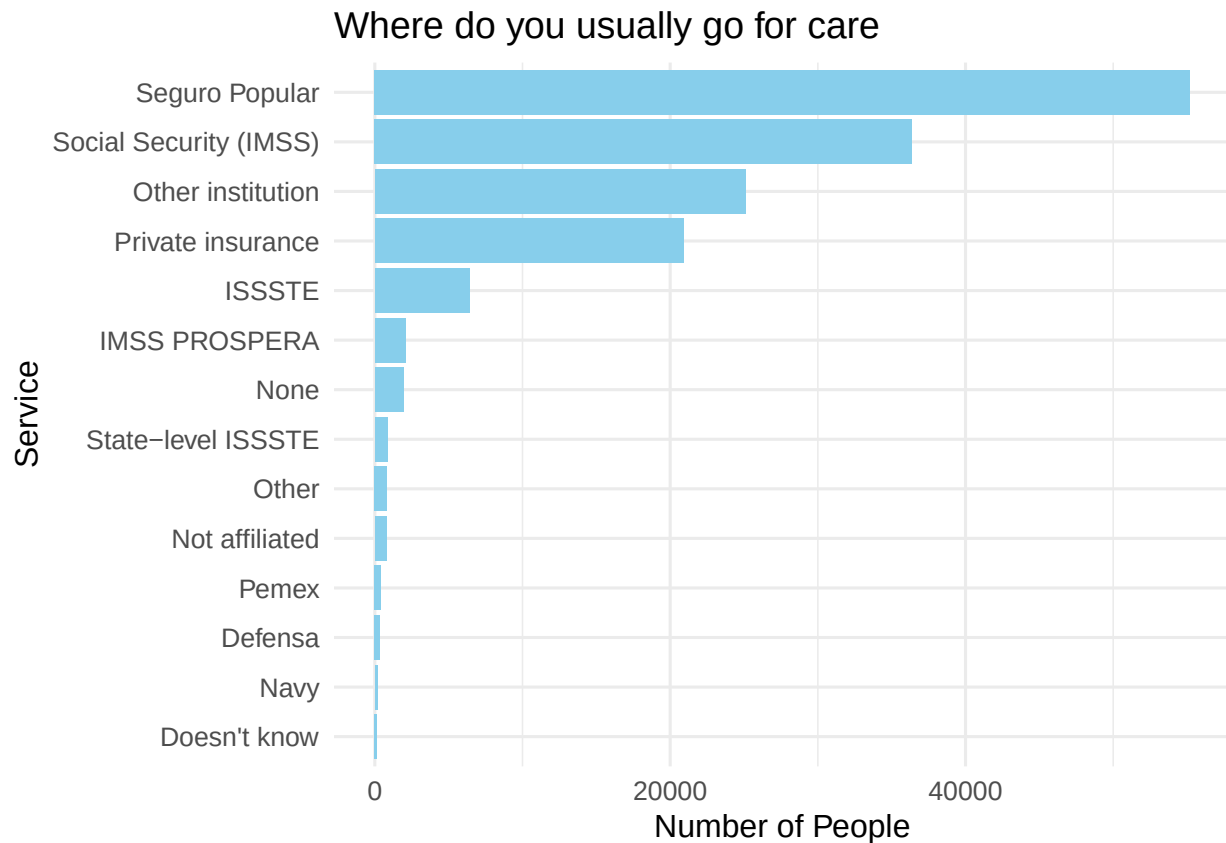
From individual_data, “do you have health problems, where do you usually go for care?”

```
labels <- c(
  "P3_9_01" = "Social Security (IMSS)",
  "P3_9_02" = "ISSSTE",
  "P3_9_03" = "State-level ISSSTE",
  "P3_9_04" = "Pemex",
  "P3_9_05" = "Defensa",
  "P3_9_06" = "Navy",
  "P3_9_07" = "Seguro Popular",
  "P3_9_08" = "IMSS PROSPERA",
  "P3_9_09" = "Private insurance",
  "P3_9_10" = "Other institution",
  "P3_9_11" = "Not affiliated",
  "P3_9_77" = "Other",
  "P3_9_12" = "None",
  "P3_9_99" = "Doesn't know"
)

# Select the columns of interest
binary_vars <- c("P3_9_01", "P3_9_02", "P3_9_03", "P3_9_04", "P3_9_05",
  "P3_9_06", "P3_9_07", "P3_9_08", "P3_9_09", "P3_9_10",
  "P3_9_11", "P3_9_77", "P3_9_12", "P3_9_99")

# Summarize counts of 1s
counts_df <- individual_data %>%
  select(all_of(binary_vars)) %>%
  summarise(across(everything(), ~ sum(. == 1, na.rm = TRUE))) %>%
  pivot_longer(cols = everything(), names_to = "code", values_to = "count") %>%
  mutate(label = labels[code]) %>%
  arrange(desc(count))

ggplot(counts_df, aes(x = reorder(label, count), y = count)) +
  geom_col(fill = "skyblue") +
  coord_flip() + # horizontal for long labels
  labs(
    title = "Where do you usually go for care",
    x = "Service",
    y = "Number of People"
  ) +
  theme_minimal(base_size = 12)
```



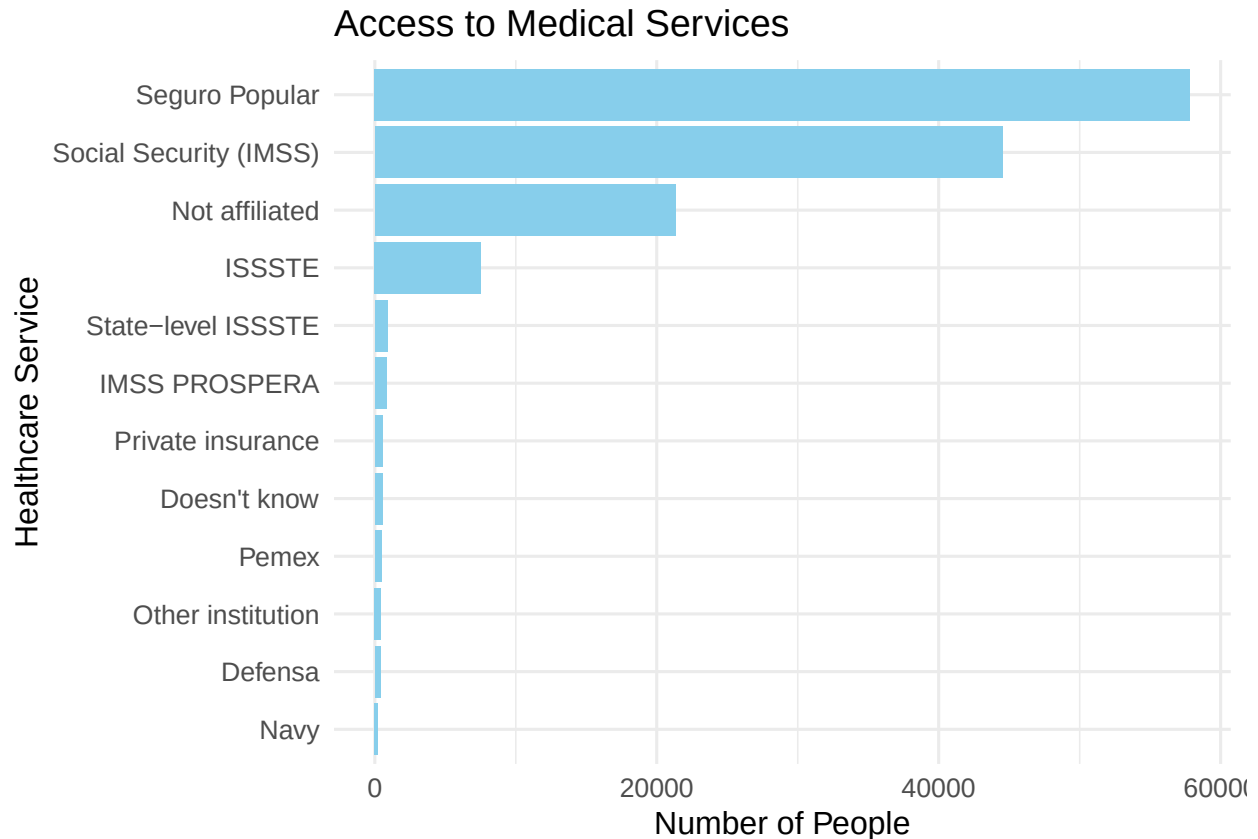
People that is entitled to or has access to these medical services

```
labels <- c(
  "1" = "Social Security (IMSS)",
  "2" = "ISSSTE",
  "3" = "State-level ISSSTE",
  "4" = "Pemex",
  "5" = "Defensa",
  "6" = "Navy",
  "7" = "Seguro Popular",
  "8" = "IMSS PROSPERA",
  "9" = "Private insurance",
  "10" = "Other institution",
  "11" = "Not affiliated",
  "99" = "Doesn't know"
)

# Create data frame with counts
access_df <- individual_data %>%
  count(P3_10_OPC1) %>%
  filter(P3_10_OPC1 %in% names(labels)) %>%
  mutate(service = labels[as.character(P3_10_OPC1)]) %>%
  arrange(desc(n)) # sort by frequency

# Plot
ggplot(access_df, aes(x = reorder(service, n), y = n)) +
  geom_col(fill = "skyblue") +
```

```
coord_flip() + # horizontal bars = better for long labels
labs(
  title = "Access to Medical Services",
  x = "Healthcare Service",
  y = "Number of People"
) +
theme_minimal(base_size = 12)
```



PROPORTION OF PEOPLE BY STRATA

Here we can see that the highest strata also goes to the highest quality health providers

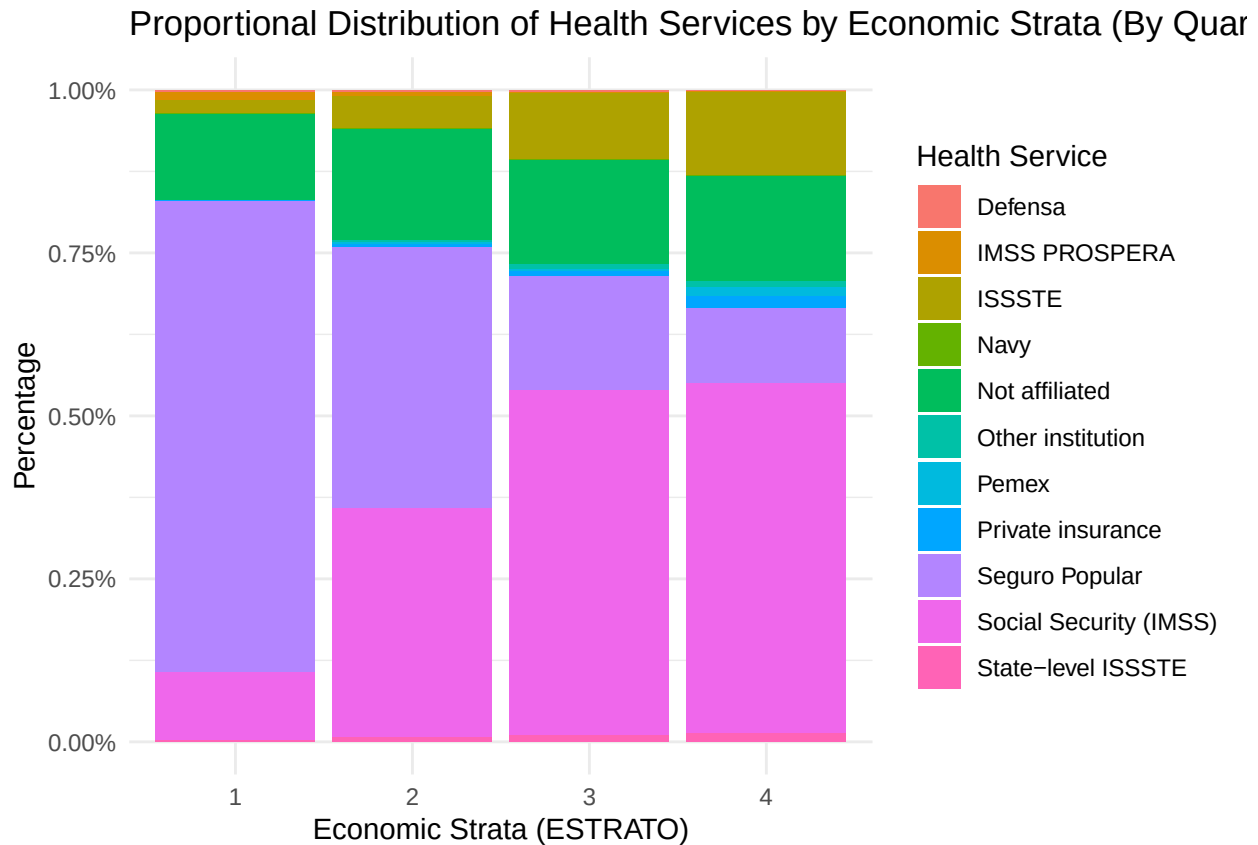
```
# Create proportional data
proportional_data <- individual_data %>%
  filter(!is.na(P3_10_OPC1) & P3_10_OPC1 != 99) %>%
  group_by(ESTRATO, P3_10_OPC1) %>%
  summarise(count = n(), .groups = "drop") %>%
  mutate(service = labels[as.character(P3_10_OPC1)]) %>%
  filter(!is.na(service)) %>%
  group_by(ESTRATO) %>%
  mutate(percentage = count / sum(count) * 100)

# Create 100% stacked bar chart
ggplot(proportional_data, aes(x = factor(ESTRATO), y = percentage, fill = service)) +
  geom_bar(stat = "identity", position = "fill") +
```

```

scale_y_continuous(labels = scales::percent_format(scale = 1)) +
labs(
  title = "Proportional Distribution of Health Services by Economic Strata (By Quartile)",
  x = "Economic Strata (ESTRATO)",
  y = "Percentage",
  fill = "Health Service"
) +
theme_minimal() +
theme(legend.position = "right")

```



CREATING QUALITY INDEX BY PROVIDER

```

# Reverse code negatively oriented Likert variables
health_quality_data <- health_quality_data %>%
  mutate(
    P4_6_rc = ifelse(P4_6 %in% 1:5, 6 - as.numeric(as.character(P4_6)), NA),
    P7_7_rc = ifelse(P7_7 %in% 1:5, 6 - as.numeric(as.character(P7_7)), NA)
  )

# Recode Yes/No variables: 1 = Yes = 1, 2 = No = 0, others = NA
yes_no_vars <- c("P7_2", "P8_8_1", "P8_8_2", "P8_8_3")
health_quality_data <- health_quality_data %>%
  mutate(across(all_of(yes_no_vars), ~ case_when(
    . == 1 ~ 1,

```

```

    . == 2 ~ 0,
    TRUE ~ NA_real_
  )))

# Convert Likert variables to numeric (1 to 5)
likert_vars <- c(
  "P3_5", "P8_2", "P8_3", "P8_4",
  "P8_5_1", "P8_5_2", "P8_5_3", "P8_5_4", "P8_5_5", "P8_5_6", "P8_6"
)
health_quality_data <- health_quality_data %>%
  mutate(across(all_of(likert_vars), ~ ifelse(. %in% 1:5, as.numeric(as.character(.)), NA_real_)))

# Define scoring rules
likert_5_vars <- c(
  "P3_5", "P7_7_rc", "P8_2", "P8_3", "P8_4",
  "P8_5_1", "P8_5_2", "P8_5_3", "P8_5_4", "P8_5_5", "P8_5_6", "P8_6"
)
binary_vars <- c("P7_2", "P8_8_1", "P8_8_2", "P8_8_3")

# Calculate raw score, max possible score, and normalized index
health_quality_data <- health_quality_data %>%
  mutate(
    quality_score_raw = rowSums(across(all_of(c(likert_5_vars, binary_vars))), na.rm = TRUE),
    quality_score_max = rowSums(across(all_of(likert_5_vars), ~ ifelse(!is.na(.), 5, 0))) +
      rowSums(across(all_of(binary_vars), ~ ifelse(!is.na(.), 1, 0))),
    quality_score = ifelse(quality_score_max > 0, quality_score_raw / quality_score_max, NA_real_)
  )

```

Compare quality score across providers.

```

# Compare quality_score across providers

# Recode provider labels
labels <- c(
  "1" = "Social Security (IMSS)",
  "2" = "ISSSTE",
  "3" = "State-level ISSSTE",
  "4" = "Pemex",
  "5" = "Defensa",
  "6" = "Navy",
  "7" = "Seguro Popular",
  "8" = "IMSS PROSPERA",
  "9" = "Private insurance",
  "10" = "Other institution",
  "11" = "Not affiliated",
  "99" = "Doesn't know"
)

health_quality_data <- health_quality_data %>%
  mutate(P3_7_label = recode(as.character(P3_7), !!!labels))

# Summarize normalized quality_score by provider
provider_summary <- health_quality_data %>%
  filter(!is.na(quality_score), !is.na(P3_7_label)) %>%

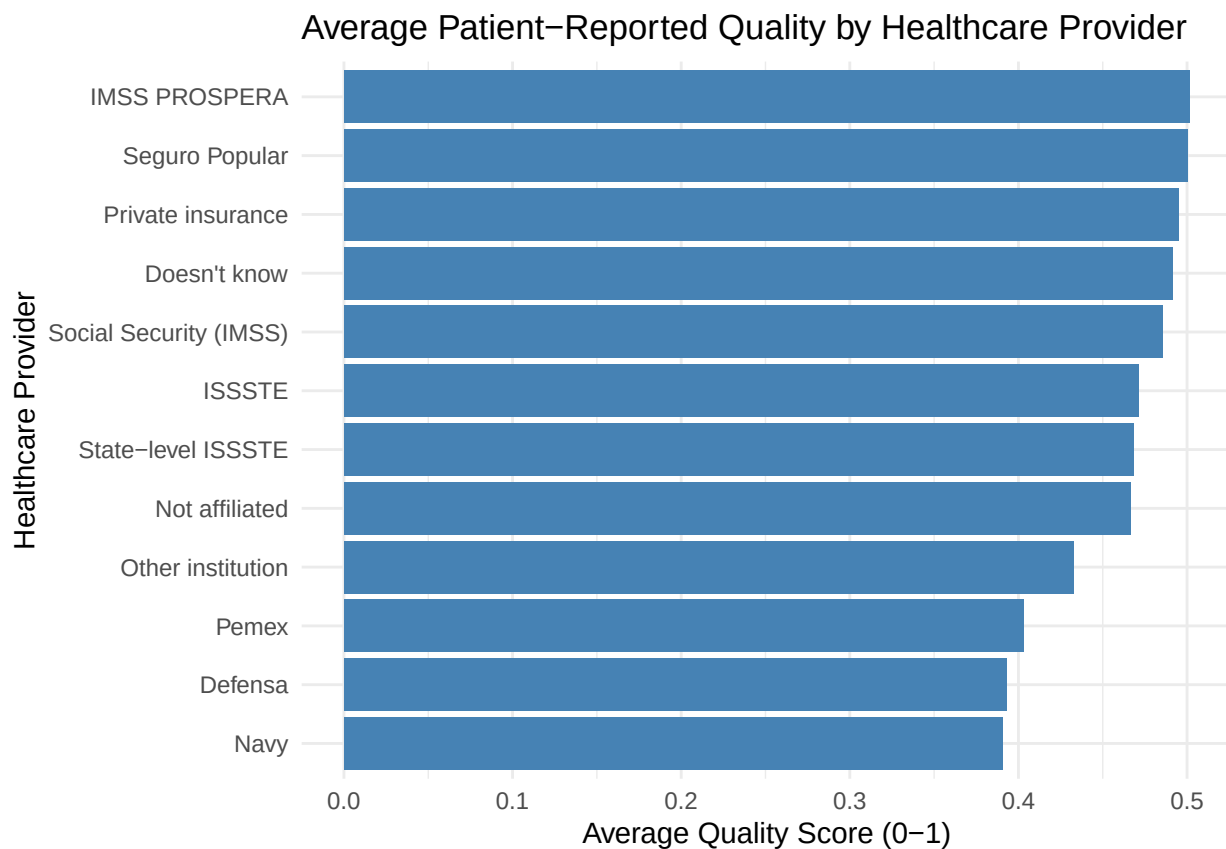
```

```

group_by(P3_7_label) %>%
  summarise(
    avg_quality = mean(quality_score, na.rm = TRUE),
    n = n()
  ) %>%
  arrange(desc(avg_quality))

# Plot average quality score by provider
ggplot(provider_summary, aes(x = fct_reorder(P3_7_label, avg_quality), y = avg_quality)) +
  geom_col(fill = "steelblue") +
  coord_flip() +
  labs(
    x = "Healthcare Provider",
    y = "Average Quality Score (0-1)",
    title = "Average Patient-Reported Quality by Healthcare Provider"
  ) +
  theme_minimal()

```



Heatmap.

```

# Labels for heatmap dimensions
dimension_labels <- c(
  P3_5 = "Quality of service",
  P4_6_rc = "Waiting time",
  P7_7_rc = "Facility conditions",
  P8_2 = "Ease of getting an appointment",

```

```

P8_3    = "Access outside regular hours",
P8_4    = "Phone access to doctor",
P8_5_1  = "Knows your history",
P8_5_2  = "Lets you ask questions",
P8_5_3  = "Spends enough time",
P8_5_4  = "Involves you in decisions",
P8_5_5  = "Explains clearly",
P8_5_6  = "Coordinates specialist care",
P8_6    = "Overall care rating"
)

# Transform to long format and label dimensions
long_data <- health_quality_data %>%
  select(P3_7_label, all_of(names(dimension_labels))) %>%
  pivot_longer(-P3_7_label, names_to = "dimension", values_to = "value") %>%
  mutate(dimension = recode(dimension, !!!dimension_labels)) %>%
  group_by(P3_7_label, dimension) %>%
  summarise(avg_val = mean(value, na.rm = TRUE), .groups = "drop") %>%
  drop_na()

# Plot heatmap with labels
ggplot(long_data, aes(x = dimension, y = fct_rev(P3_7_label), fill = avg_val)) +
  geom_tile(color = "white") +
  scale_fill_viridis_c(limits = c(0, 5)) +
  labs(
    x = "Quality Dimension",
    y = "Healthcare Provider",
    fill = "Avg. Score",
    title = "Patient Experience Scores by Provider and Dimension"
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1),
    axis.text = element_text(size = 10),
    plot.title = element_text(face = "bold")
  )

```

Patient Experience Scores by Provider and Dimension



RETURN PROPENSITY

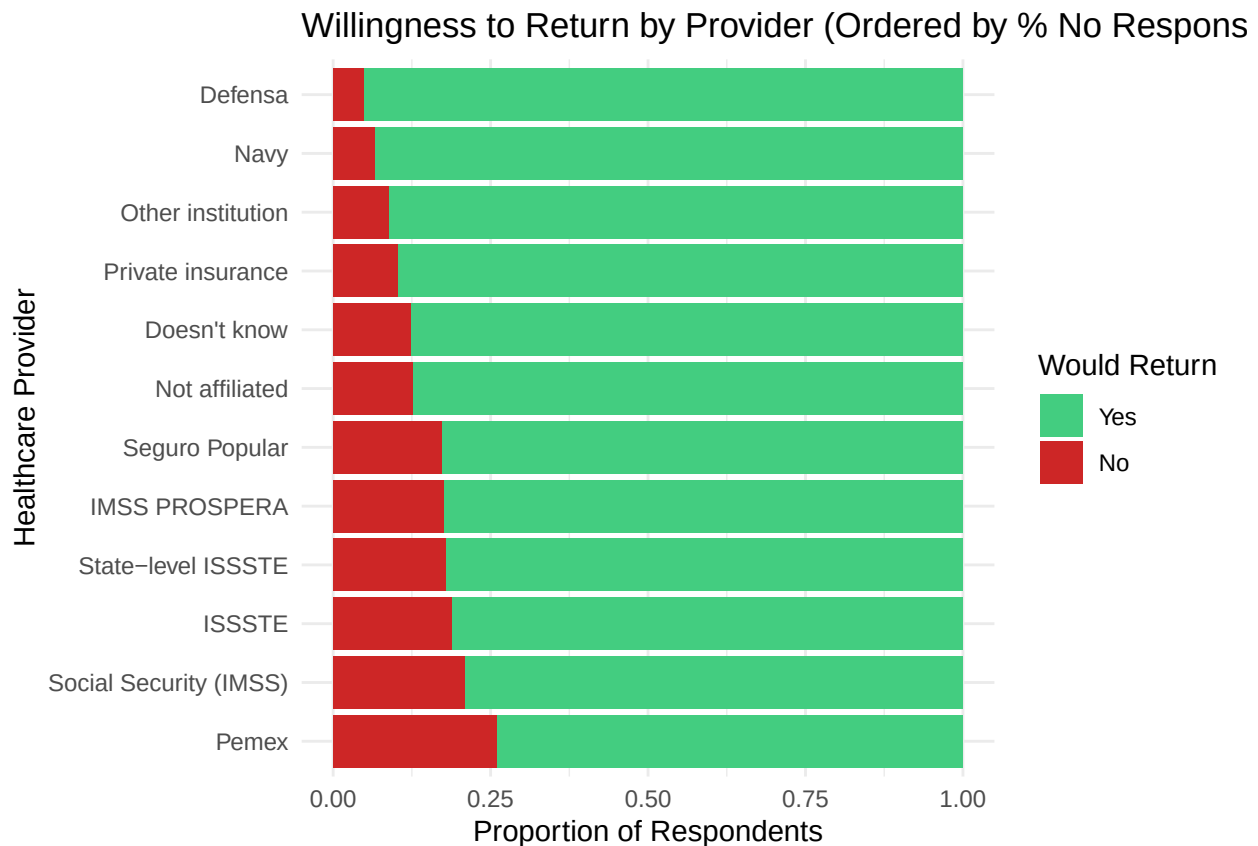
```
# Calculate return proportions
return_prop <- health_quality_data %>%
  filter(!is.na(P7_2), !is.na(P3_7_label)) %>%
  group_by(P3_7_label, P7_2) %>%
  summarise(n = n(), .groups = "drop") %>%
  group_by(P3_7_label) %>%
  mutate(prop = n / sum(n)) %>%
  ungroup()

# Extract % No (P7_2 == 0) per provider
no_order <- return_prop %>%
  filter(P7_2 == 0) %>%
  select(P3_7_label, no_prop = prop)

# Reorder factor levels based on descending % No
ordered_labels <- no_order %>%
  arrange(desc(no_prop)) %>%
  pull(P3_7_label)

# Apply factor levels to full data
return_prop <- return_prop %>%
  mutate(P3_7_label = factor(P3_7_label, levels = ordered_labels))
```

```
# Plot
ggplot(return_prop, aes(x = P3_7_label, y = prop, fill = factor(P7_2, levels = c(1, 0)))) +
  geom_col(position = "fill") +
  coord_flip() +
  scale_fill_manual(values = c("1" = "seagreen3", "0" = "firebrick3"), labels = c("1" = "Yes", "0" = "No")) +
  labs(
    x = "Healthcare Provider",
    y = "Proportion of Respondents",
    fill = "Would Return",
    title = "Willingness to Return by Provider (Ordered by % No Responses)"
  ) +
  theme_minimal()
```



```
# Reasons
# Create named labels for reasons
yes_labels <- c(
  P7_3_1 = "No other option",
  P7_3_2 = "Treated well",
  P7_3_3 = "Good care",
  P7_3_4 = "No payment",
  P7_3_5 = "Close to home",
  P7_3_6 = "Affordable",
  P7_3_7 = "Quick attention",
  P7_3_8 = "Received meds",
  P7_3_9 = "Short wait",
  P7_3_10 = "Clean facility",
)
```

```

P7_3_11 = "Doctor explained",
P7_3_12 = "Other",
P7_3_13 = "Doesn't know"
)

no_labels <- c(
  P7_4_1 = "Treated badly",
  P7_4_2 = "Disagreed with diagnosis",
  P7_4_3 = "Disagreed with treatment",
  P7_4_4 = "Did not improve",
  P7_4_5 = "Expensive",
  P7_4_6 = "Far from home",
  P7_4_7 = "No meds",
  P7_4_8 = "No materials",
  P7_4_9 = "Family not allowed",
  P7_4_10 = "Long wait",
  P7_4_11 = "Dirty facility",
  P7_4_12 = "Doctor didn't explain",
  P7_4_13 = "Other",
  P7_4_14 = "Doesn't know"
)

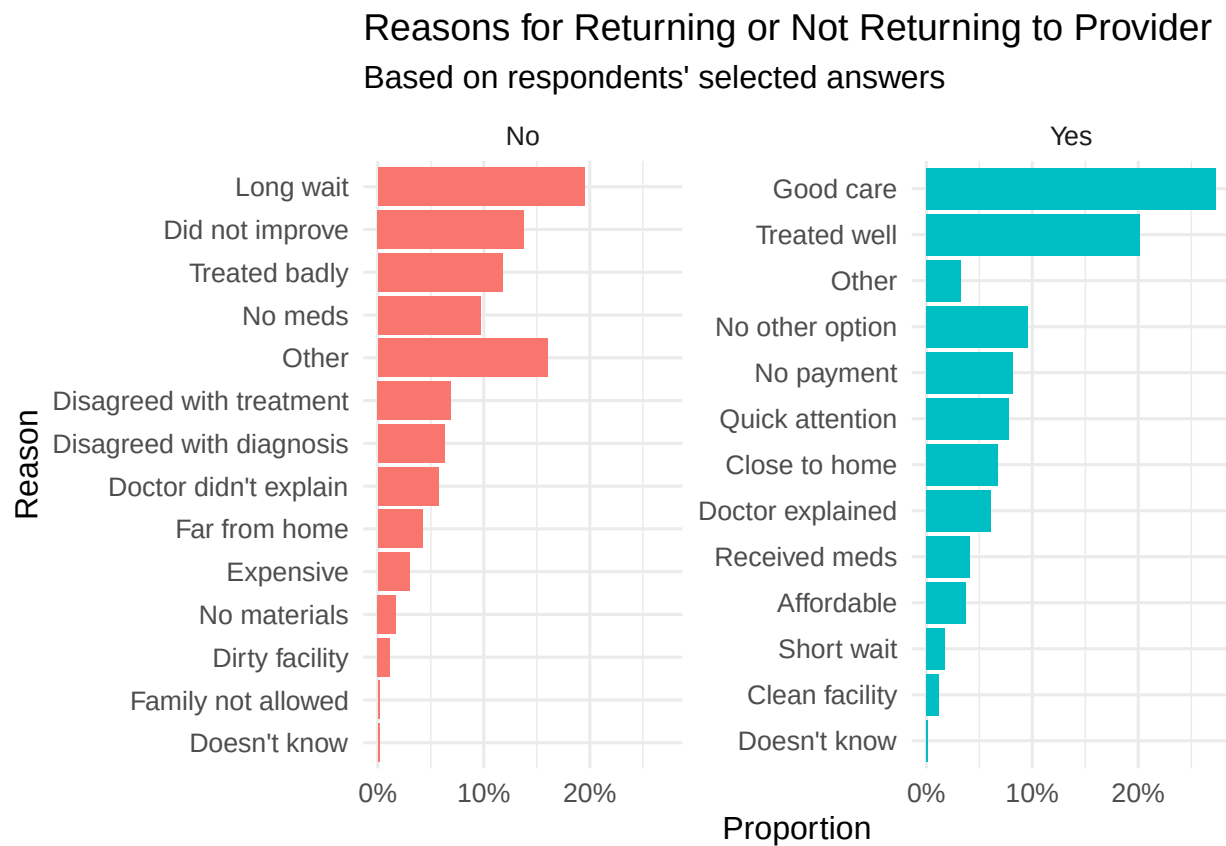
# Pivot long and tag reasons
reasons_long <- health_quality_data %>%
  select(starts_with("P7_3_"), starts_with("P7_4_")) %>%
  pivot_longer(
    everything(),
    names_to = "reason_code",
    values_to = "response"
  ) %>%
  filter(response == 1) %>% # Only keep selected reasons
  mutate(
    type = case_when(
      str_detect(reason_code, "P7_3") ~ "Yes",
      str_detect(reason_code, "P7_4") ~ "No"
    ),
    reason = case_when(
      reason_code %in% names(yes_labels) ~ yes_labels[reason_code],
      reason_code %in% names(no_labels) ~ no_labels[reason_code],
      TRUE ~ "Unknown"
    )
  )

# Summarize counts
reason_summary <- reasons_long %>%
  group_by(type, reason) %>%
  summarise(n = n(), .groups = "drop") %>%
  group_by(type) %>%
  mutate(prop = n / sum(n)) %>%
  ungroup()

# Plot
ggplot(reason_summary, aes(x = reorder(reason, prop), y = prop, fill = type)) +

```

```
geom_col(show.legend = FALSE) +
facet_wrap(~type, scales = "free_y") +
coord_flip() +
scale_y_continuous(labels = scales::percent_format()) +
labs(
  x = "Reason",
  y = "Proportion",
  title = "Reasons for Returning or Not Returning to Provider",
  subtitle = "Based on respondents' selected answers"
) +
theme_minimal(base_size = 12)
```



----- BEFORE STARTING: HEALTH CONDITIONS -----

```
# Create condition mapping
condition_labels <- c(
  "1" = "Respiratory infections",
  "2" = "Pneumonia",
  "3" = "COPD",
  "4" = "Cough/Sore throat",
  "5" = "Tuberculosis",
  "6" = "Ear infection",
  "7" = "Conjunctivitis",
  "8" = "Asthma",
  "9" = "Allergies",
  "10" = "Heart disease",
```

```

"11" = "Rheumatic fever",
"12" = "Diabetes",
"13" = "Hypertension",
"14" = "Stroke",
"15" = "Obesity",
"16" = "Arthritis",
"17" = "Diarrhea",
"18" = "Gastritis/Ulcer",
"19" = "Colitis",
"20" = "Parasites",
"21" = "Hepatitis",
"22" = "Kidney disease",
"23" = "UTI",
"24" = "Exanthematous disease",
"25" = "STI",
"26" = "HIV/AIDS",
"28" = "Dengue",
"29" = "Poisoning",
"30" = "Alcoholism",
"32" = "Drug-related illness",
"33" = "Accidental injury",
"34" = "Injury from aggression",
"36" = "Stress",
"37" = "Depression",
"38" = "Skin issues",
"39" = "Oral disease",
"40" = "Headache",
"41" = "Fever (unspecified)",
"42" = "Folk illnesses",
"43" = "Pregnancy",
"44" = "Cancer",
"45" = "Other",
"99" = "Don't know"
)

# Prepare the data - convert condition codes to factors with labels
health_quality_data <- health_quality_data %>%
  mutate(
    primary_condition_code = as.character(P1_2),
    primary_condition = factor(primary_condition_code,
                              levels = names(condition_labels),
                              labels = condition_labels)
  )

# Rename provider
health_quality_data <- health_quality_data %>%
  mutate(provider = P3_7)

# Step 1: Chi-square test to see if conditions differ by provider
chi_result <- chisq.test(table(health_quality_data$primary_condition, health_quality_data$provider))

## Warning in chisq.test(table(health_quality_data$primary_condition,
## health_quality_data$provider)): Chi-squared approximation may be incorrect

```

```
print(paste("Chi-square test result: X2 =", round(chi_result$statistic, 2),
            ", df =", chi_result$parameter,
            ", p-value =", format.pval(chi_result$p.value, digits = 3)))
```

```
## [1] "Chi-square test result: X2 = 1801.5 , df = 462 , p-value = <2e-16"
```

```
# Step 2: Calculate condition frequencies by provider
condition_by_provider <- health_quality_data %>%
  group_by(provider, primary_condition) %>%
  summarize(count = n(), .groups = "drop") %>%
  group_by(provider) %>%
  mutate(
    total = sum(count),
    percentage = count / total * 100
  ) %>%
  arrange(provider, desc(percentage))

# Step 3: Get the top 5 conditions for each provider
top_conditions_by_provider <- condition_by_provider %>%
  group_by(provider) %>%
  slice_max(order_by = percentage, n = 5) %>%
  ungroup()

# Step 4: Find conditions with the biggest variation across providers
condition_variation <- condition_by_provider %>%
  group_by(primary_condition) %>%
  summarize(
    max_pct = max(percentage, na.rm = TRUE),
    min_pct = min(percentage, na.rm = TRUE),
    variation = max_pct - min_pct,
    avg_pct = mean(percentage, na.rm = TRUE)
  ) %>%
  filter(!is.na(primary_condition)) %>%
  arrange(desc(variation))

top_varying_conditions <- condition_variation %>%
  slice_max(order_by = variation, n = 10) %>%
  pull(primary_condition)

# Step 5: Visualize the results

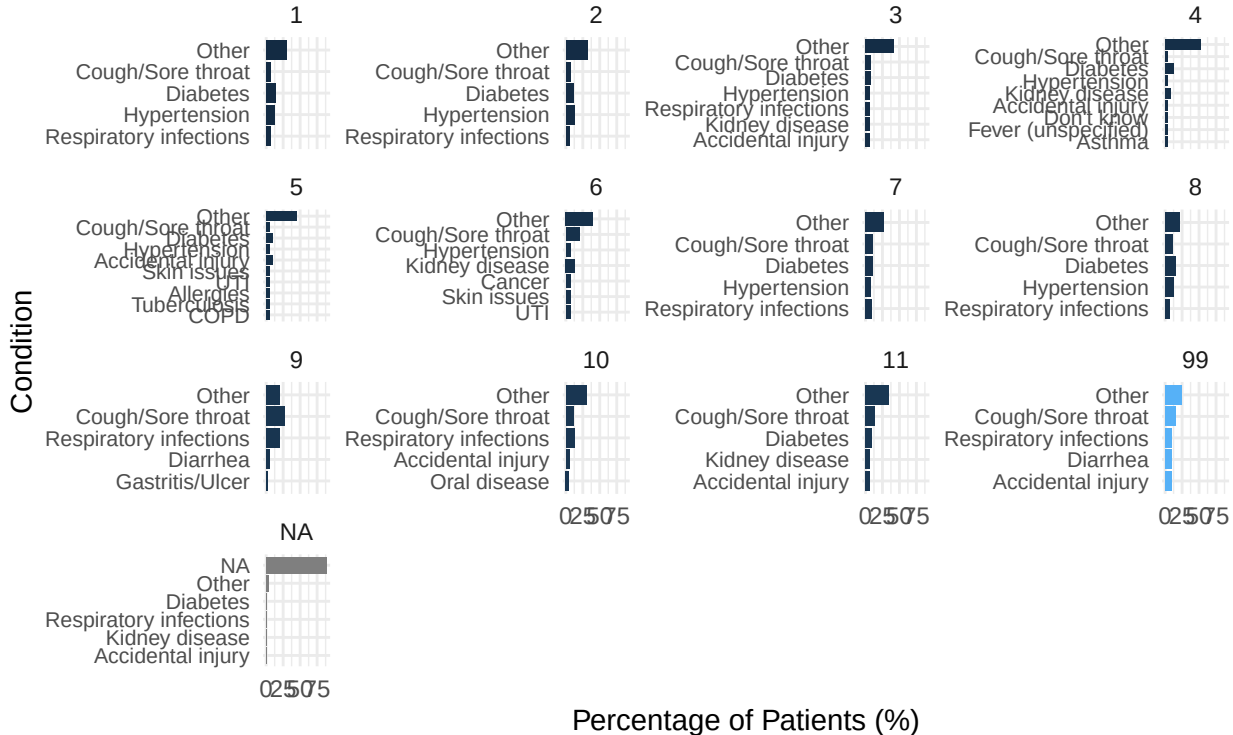
# Plot 1: Top 5 conditions by provider
plot1 <- ggplot(top_conditions_by_provider,
  aes(x = reorder(primary_condition, percentage), y = percentage, fill = provider)) +
  geom_col() +
  facet_wrap(~ provider, scales = "free_y") +
  coord_flip() +
  theme_minimal() +
  labs(
    title = "Top 5 Primary Conditions by Provider Type",
    subtitle = "Percentage of patients seeking care for each condition",
    x = "Condition",
    y = "Percentage of Patients (%)"
  )
```

```
) +
theme(
  legend.position = "none",
  axis.text.y = element_text(size = 8)
)

print(plot1)
```

Top 5 Primary Conditions by Provider Type

Percentage of patients seeking care for each condition



Hypothesis ## Hypo 1 Without capitation or fee-for-service incentives and with low monitoring, providers may induce less services.

```
# Label providers
health_quality_data <- health_quality_data %>%
  mutate(
    provider = factor(provider, levels = c(1,2,3,4,5,6,7,8,9, 10, 11),
                      labels = c("IMSS", "ISSSTE", "State-level ISSSTE",
                                "Pemex", "Defensa", "Navy", "Seguro Popular",
                                "IMSS PROSPERA", "Other", "Private", "Other")),
    test_requested = ifelse(P6_1 == 1, 1, 0),
    sex = factor(SEX0, labels = c("Male", "Female")),
    medications = as.numeric(as.character(P5_1)),
    condition = as.factor(P1_2), # self-reported condition
    age = as.numeric(EDAD)
  )

health_quality_data$provider <- relevel(health_quality_data$provider, ref = "Private")
```

```

# Diagnostic tests requested
model_tests <- lm(test_requested ~ provider + age + sex + condition,
                  data = health_quality_data)

# Number of medications prescribed
model_meds <- lm(medications ~ provider + age + sex + condition,
                 data = health_quality_data)

# Summary with robust standard errors
coeftest(model_tests, vcov = vcovHC(model_tests, type = "HC1"))

## Warning in meatHC(x, type = type, omega = omega): HC1 covariances become (close
## to) singular if hat values are (close to) 1 as for observation(s) 318, 4061

##
## t test of coefficients:
##
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)    0.08957877  0.01645123   5.4451 5.359e-08 ***
## providerIMSS    0.05044370  0.01633948   3.0872 0.0020285 **
## providerISSSTE   0.08517283  0.02736824   3.1121 0.0018653 **
## providerState-level ISSSTE 0.18325876  0.06789343   2.6992 0.0069676 **
## providerPemex    0.06962936  0.10412740   0.6687 0.5037135
## providerDefensa  0.14145602  0.10326516   1.3698 0.1707844
## providerNavy     -0.03131052  0.11240666  -0.2785 0.7806011
## providerSeguro Popular -0.00614501  0.01537318  -0.3997 0.6893733
## providerIMSS PROSPERA -0.06953005  0.04445736  -1.5640 0.1178709
## providerOther    -0.08396618  0.01529770  -5.4888 4.193e-08 ***
## age              0.00082821  0.00025097   3.3000 0.0009719 ***
## sexFemale        0.02279731  0.01111107   2.0518 0.0402307 *
## condition2        0.31332122  0.09173008   3.4157 0.0006399 ***
## condition3        0.15573164  0.06581438   2.3662 0.0179984 *
## condition4       -0.04365731  0.01383779  -3.1549 0.0016124 **
## condition5        0.79616833  0.07422829  10.7259 < 2.2e-16 ***
## condition6        0.05655850  0.05409654   1.0455 0.2958243
## condition7        0.04637823  0.07061276   0.6568 0.5113339
## condition8        0.23000721  0.06321591   3.6384 0.0002763 ***
## condition9        0.09307046  0.04001861   2.3257 0.0200650 *
## condition10       0.47774479  0.05706398   8.3721 < 2.2e-16 ***
## condition11       0.33043306  0.11508078   2.8713 0.0041003 **
## condition12       0.27011595  0.02597048  10.4009 < 2.2e-16 ***
## condition13       0.20092204  0.02754422   7.2945 3.338e-13 ***
## condition14       0.46664279  0.14155344   3.2966 0.0009837 ***
## condition15       0.29981880  0.16311276   1.8381 0.0660902 .
## condition16       0.22584554  0.06519845   3.4640 0.0005356 ***
## condition17       0.03092089  0.02642800   1.1700 0.2420401
## condition18       0.21995037  0.04188934   5.2507 1.561e-07 ***
## condition19       0.27400725  0.06170202   4.4408 9.104e-06 ***
## condition20       0.16920234  0.07440426   2.2741 0.0229914 *
## condition21       0.90276323  0.03508918  25.7277 < 2.2e-16 ***
## condition22       0.41604254  0.04720470   8.8136 < 2.2e-16 ***
## condition23       0.38194691  0.04612751   8.2802 < 2.2e-16 ***
## condition24       0.00201301  0.10106306   0.0199 0.9841091

```

```
## condition25      0.24489752  0.15854734  1.5446 0.1224815
## condition26      0.78748787  0.01936271 40.6703 < 2.2e-16 ***
## condition28      0.46268817  0.11421755  4.0509 5.158e-05 ***
## condition29     -0.02668093  0.10663356 -0.2502 0.8024314
## condition30     -0.11490558  0.01743331 -6.5912 4.692e-11 ***
## condition32      0.83306084  0.01790697 46.5216 < 2.2e-16 ***
## condition33      0.35974173  0.03071409 11.7126 < 2.2e-16 ***
## condition34      0.53698657  0.10679903  5.0280 5.083e-07 ***
## condition36      0.27404224  0.08376007  3.2718 0.0010742 **
## condition37      0.32241378  0.09556237  3.3739 0.0007454 ***
## condition38      0.09973604  0.05645263  1.7667 0.0773199 .
## condition39      0.01753124  0.03255590  0.5385 0.5902519
## condition40      0.27091541  0.05799675  4.6712 3.052e-06 ***
## condition41      0.09569169  0.06633425  1.4426 0.1491882
## condition42      0.23297972  0.18392748  1.2667 0.2053084
## condition43      0.51345807  0.03708619 13.8450 < 2.2e-16 ***
## condition44      0.44848362  0.04955129  9.0509 < 2.2e-16 ***
## condition45      0.27611850  0.01706936 16.1763 < 2.2e-16 ***
## condition99      0.26931763  0.15267972  1.7639 0.0777873 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
coeftest(model_meds, vcov = vcovHC(model_meds, type = "HC1"))
```

```
## Warning in meatHC(x, type = type, omega = omega): HC1 covariances become (close
## to) singular if hat values are (close to) 1 as for observation(s) 318, 4061
```

```
##
## t test of coefficients:
##
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    2.9388124  0.0861955 34.0947 < 2.2e-16 ***
## providerIMSS     0.1608057  0.0813199  1.9774 0.0480317 *
## providerISSSTE  -0.2635983  0.1227776 -2.1470 0.0318320 *
## providerState-level ISSSTE  0.2027887  0.3037149  0.6677 0.5043514
## providerPemex   -0.7266832  0.3173313 -2.2900 0.0220528 *
## providerDefensa -0.9902343  0.2898606 -3.4162 0.0006386 ***
## providerNavy    -0.6289106  0.3082495 -2.0403 0.0413626 *
## providerSeguro Popular -0.4104891  0.0730577 -5.6187 2.000e-08 ***
## providerIMSS PROSPERA -0.8198335  0.1473560 -5.5636 2.743e-08 ***
## providerOther   -0.0096453  0.0866662 -0.1113 0.9113879
## age             0.0082047  0.0010270  7.9889 1.588e-15 ***
## sexFemale       0.0101051  0.0547569  0.1845 0.8535914
## condition2     -0.3414622  0.2658007 -1.2847 0.1989567
## condition3      0.4585308  0.2297161  1.9961 0.0459655 *
## condition4     -0.1273459  0.0683256 -1.8638 0.0623914 .
## condition5      0.9077985  0.2668052  3.4025 0.0006716 ***
## condition6     -0.4756685  0.2558433 -1.8592 0.0630394 .
## condition7     -1.4963977  0.2212471 -6.7635 1.460e-11 ***
## condition8      0.6620213  0.2554767  2.5913 0.0095814 **
## condition9     -0.4798386  0.1414126 -3.3932 0.0006948 ***
## condition10    -0.0383723  0.3309385 -0.1159 0.9076956
## condition11    -0.1135623  0.4464309 -0.2544 0.7992110
```

```
## condition12      0.4519637  0.1111841   4.0650 4.857e-05 ***
## condition13      0.0085387  0.1078464   0.0792 0.9368963
## condition14      0.7523484  0.4512593   1.6672 0.0955168 .
## condition15     -1.9716976  0.3461097  -5.6967 1.272e-08 ***
## condition16      0.9401270  0.3826038   2.4572 0.0140280 *
## condition17     -0.1813625  0.0974731  -1.8606 0.0628380 .
## condition18     -0.3292306  0.1299829  -2.5329 0.0113354 *
## condition19     -0.0144916  0.2003535  -0.0723 0.9423413
## condition20     -0.2720116  0.2386973  -1.1396 0.2545068
## condition21     -1.1227061  0.4689249  -2.3942 0.0166830 *
## condition22     -0.1350100  0.2088200  -0.6465 0.5179531
## condition23     -0.4100191  0.1476177  -2.7776 0.0054917 **
## condition24      0.5755115  0.7249679   0.7938 0.4273140
## condition25     -1.4402086  0.3547015  -4.0603 4.955e-05 ***
## condition26     -3.6020031  0.0834955 -43.1401 < 2.2e-16 ***
## condition28     -0.5938752  0.3174874  -1.8705 0.0614509 .
## condition29     -1.8770839  0.3807932  -4.9294 8.444e-07 ***
## condition30     -1.8401006  0.0714185 -25.7651 < 2.2e-16 ***
## condition32     -0.3662697  0.0770143  -4.7559 2.017e-06 ***
## condition33     -1.1303817  0.0922415 -12.2546 < 2.2e-16 ***
## condition34     -0.7896383  0.3601169  -2.1927 0.0283608 *
## condition36     -0.9060168  0.1895938  -4.7787 1.801e-06 ***
## condition37     -0.0434405  0.3587165  -0.1211 0.9036155
## condition38     -0.4360098  0.2342915  -1.8610 0.0627913 .
## condition39     -1.7116993  0.1189313 -14.3923 < 2.2e-16 ***
## condition40     -0.7955059  0.1799392  -4.4210 9.979e-06 ***
## condition41     -0.4129954  0.2124184  -1.9443 0.0519060 .
## condition42     -0.5017538  0.3963706  -1.2659 0.2056029
## condition43     -1.3171075  0.1181954 -11.1435 < 2.2e-16 ***
## condition44     -1.3571965  0.2122272  -6.3950 1.712e-10 ***
## condition45     -0.7311641  0.0838186  -8.7232 < 2.2e-16 ***
## condition99     -0.9707624  0.5690591  -1.7059 0.0880710 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Health conditions

```
# -- FOR HEALTH CONDITIONS --
condition_map <- data.frame(
  term = paste0("condition", c(
    2, 3, 4, 5, 6, 7, 8, 9, 10, 11,
    12, 13, 14, 15, 16, 17, 18, 19, 20, 21,
    22, 23, 24, 25, 26, 28, 29, 30, 32, 33,
    34, 36, 37, 38, 39, 40, 41, 42, 43, 44,
    45, 99)
),
  condition = c(
    "Pneumonia", "COPD", "Cold/Sore throat", "Tuberculosis", "Ear infection", "Conjunctivitis", "Asthma",
    "Rheumatic fever", "Diabetes", "Hypertension", "Stroke", "Obesity", "Arthritis", "Diarrhea", "Gastritis",
    "Hepatitis", "Kidney disease", "UTI", "Exanthematous disease", "STI", "HIV/AIDS", "Dengue", "Poisoning",
    "Accidental injury", "Injury from aggression", "Stress", "Depression", "Skin issues", "Oral diseases",
    "Folk illnesses", "Pregnancy", "Cancer", "Other", "Don't know"
)
)
```

```

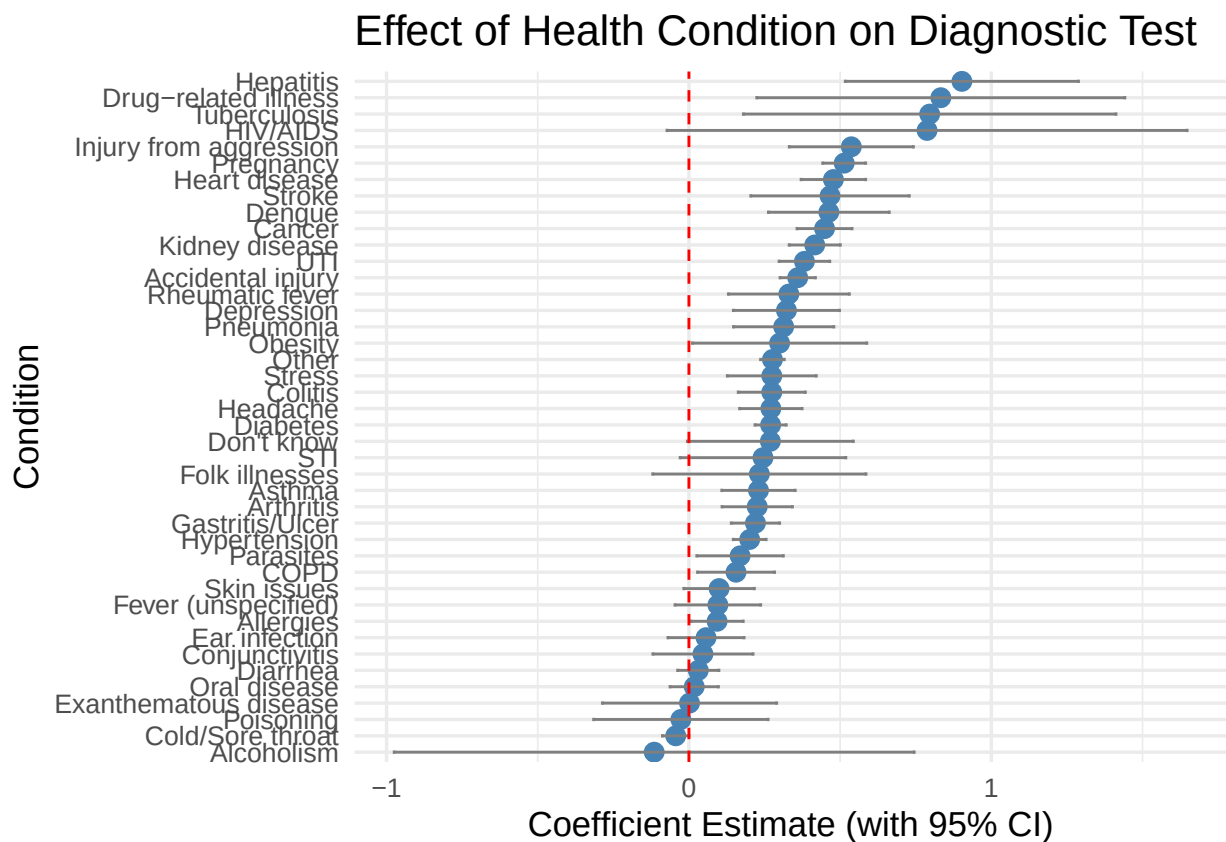
tidy_model <- tidy(model_tests, conf.int = TRUE)

# Filter only condition variables
tidy_conditions <- tidy_model %>%
  filter(grepl("^condition\\d+", term)) %>%
  left_join(condition_map, by = "term") %>%
  arrange(estimate)

# Create factor with ordering for plotting
tidy_conditions$condition <- factor(tidy_conditions$condition, levels = tidy_conditions$condition)

ggplot(tidy_conditions, aes(x = estimate, y = condition)) +
  geom_point(color = "steelblue", size = 3) +
  geom_errorbarh(aes(xmin = conf.low, xmax = conf.high), height = 0.2, color = "gray50") +
  geom_vline(xintercept = 0, linetype = "dashed", color = "red") +
  labs(
    title = "Effect of Health Condition on Diagnostic Test",
    x = "Coefficient Estimate (with 95% CI)",
    y = "Condition"
  ) +
  theme_minimal(base_size = 13)

```



```

# -- NOW FOR PROVIDERS--
provider_map <- data.frame(
  term = c(

```

```

    "providerIMSS", "providerISSSTE", "providerState-level ISSSTE",
    "providerPemex", "providerDefensa",
    "providerNavy", "providerSeguro Popular", "providerIMSS PROSPERA",
    "providerOther"
  ),
  provider = c(
    "Social Security (IMSS)", "ISSSTE", "State-level ISSSTE", "Pemex", "Defensa",
    "Navy", "Seguro Popular", "IMSS PROSPERA", "Other institution"
  )
)

# Tidy your model
tidy_model <- tidy(model_tests, conf.int = TRUE)

# Filter for provider terms only
tidy_providers <- tidy_model %>%
  filter(term %in% provider_map$term) %>%
  left_join(provider_map, by = "term") %>%
  arrange(estimate)

# Join the results on quality by provider
provider_summary <- provider_summary %>%
  rename(provider = P3_7_label)

tidy_providers <- tidy_providers %>%
  left_join(provider_summary, by = "provider")

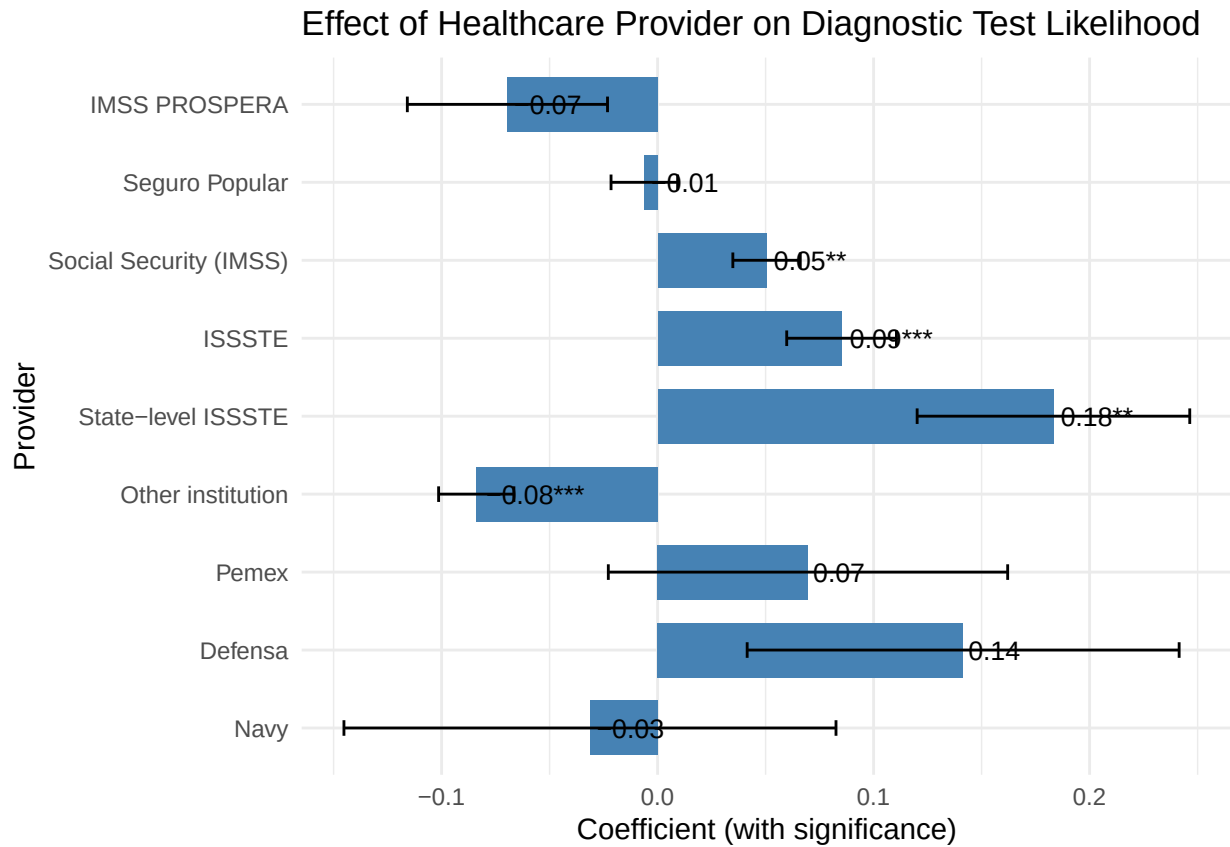
# 1. Add significance stars
tidy_providers <- tidy_providers %>%
  mutate(
    stars = case_when(
      p.value < 0.001 ~ "***",
      p.value < 0.01 ~ "**",
      p.value < 0.05 ~ "*",
      p.value < 0.1 ~ ".",
      TRUE ~ ""
    ),
    coef_label = paste0(sprintf("%.2f", estimate), stars)
  )

# 2. Order by average quality (ensure provider is factor)
tidy_providers <- tidy_providers %>%
  mutate(provider = fct_reorder(provider, avg_quality))

# 3. Plot
ggplot(tidy_providers, aes(x = provider, y = estimate)) +
  geom_col(fill = "steelblue", width = 0.7) +
  geom_errorbar(aes(ymin = estimate - std.error,
                    ymax = estimate + std.error), width = 0.2) +
  geom_text(aes(label = coef_label), hjust = -0.1, size = 3.5) +
  coord_flip() +
  labs(
    title = "Effect of Healthcare Provider on Diagnostic Test Likelihood",

```

```
x = "Provider",
y = "Coefficient (with significance)"
) +
theme_minimal()
```



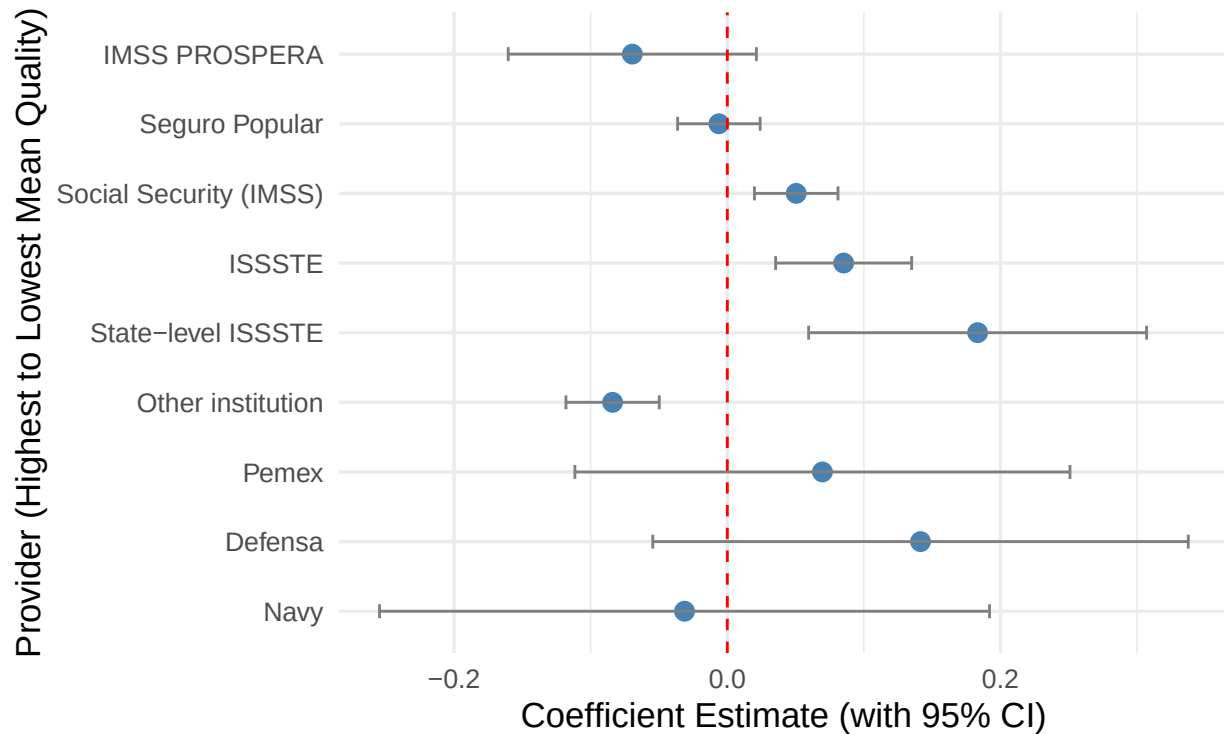
```
plot_model(model_tests, show.values = TRUE, value.offset = 0.3) +
coord_flip()
```

```
## Coordinate system already present. Adding new coordinate system, which will
## replace the existing one.
```



```
ggplot(tidy_providers, aes(x = estimate, y = provider)) +
  geom_point(color = "steelblue", size = 3) +
  geom_errorbarh(aes(xmin = conf.low, xmax = conf.high), height = 0.2, color = "gray50") +
  geom_vline(xintercept = 0, linetype = "dashed", color = "red") +
  labs(
    title = "Effect of Provider Type on Number of Diagnostic Tests",
    subtitle = "Ordered by Mean Quality of Care",
    x = "Coefficient Estimate (with 95% CI)",
    y = "Provider (Highest to Lowest Mean Quality)"
  ) +
  theme_minimal(base_size = 13)
```

Effect of Provider Type on Number of Diagnostic Ordered by Mean Quality of Care



```
#####
##### PLOTS FOR MODEL_MEDS
#####

# -- FOR HEALTH CONDITIONS --
condition_map <- data.frame(
  term = paste0("condition", c(
    2, 3, 4, 5, 6, 7, 8, 9, 10, 11,
    12, 13, 14, 15, 16, 17, 18, 19, 20, 21,
    22, 23, 24, 25, 26, 28, 29, 30, 32, 33,
    34, 36, 37, 38, 39, 40, 41, 42, 43, 44,
    45, 99)
),
  condition = c(
    "Pneumonia", "COPD", "Cold/Sore throat", "Tuberculosis", "Ear infection", "Conjunctivitis", "Asthma",
    "Rheumatic fever", "Diabetes", "Hypertension", "Stroke", "Obesity", "Arthritis", "Diarrhea", "Gastr",
    "Hepatitis", "Kidney disease", "UTI", "Exanthematous disease", "STI", "HIV/AIDS", "Dengue", "Poison",
    "Accidental injury", "Injury from aggression", "Stress", "Depression", "Skin issues", "Oral disease",
    "Folk illnesses", "Pregnancy", "Cancer", "Other", "Don't know"
  )
)

tidy_model_meds <- tidy(model_meds, conf.int = TRUE)

# Filter only condition variables
tidy_conditions_meds <- tidy_model_meds %>%
  filter(grepl("^condition\\d+", term)) %>%
```

```

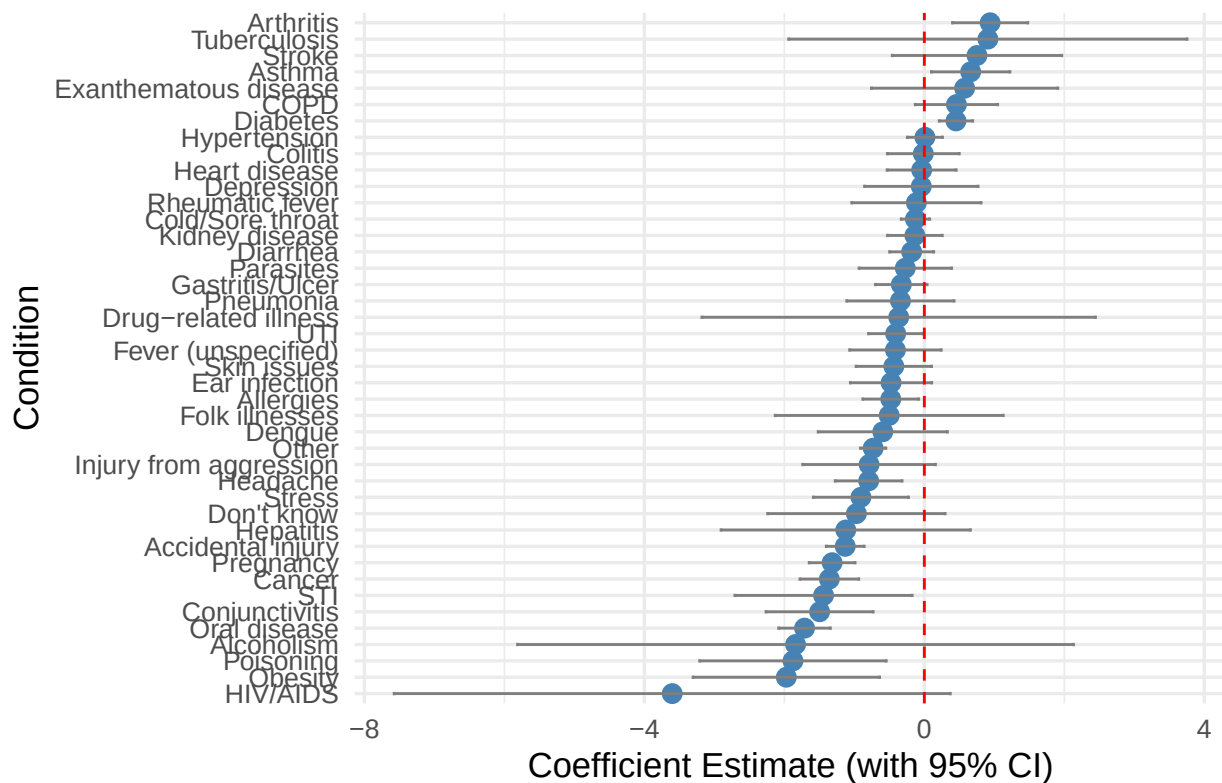
left_join(condition_map, by = "term") %>%
  arrange(estimate)

# Create factor with ordering for plotting
tidy_conditions_meds$condition <- factor(tidy_conditions_meds$condition, levels = tidy_conditions_meds$

ggplot(tidy_conditions_meds, aes(x = estimate, y = condition)) +
  geom_point(color = "steelblue", size = 3) +
  geom_errorbarh(aes(xmin = conf.low, xmax = conf.high), height = 0.2, color = "gray50") +
  geom_vline(xintercept = 0, linetype = "dashed", color = "red") +
  labs(
    title = "Effect of Health Condition on Meds Prescription",
    x = "Coefficient Estimate (with 95% CI)",
    y = "Condition"
  ) +
  theme_minimal(base_size = 13)

```

Effect of Health Condition on Meds Prescription



```

# -- NOW FOR PROVIDERS--
provider_map <- data.frame(
  term = c(
    "providerIMSS", "providerISSSTE", "providerState-level ISSSTE",
    "providerPemex", "providerDefensa",
    "providerNavy", "providerSeguro Popular", "providerIMSS PROSPERA",
    "providerOther"
  )
),

```

```

provider = c(
  "Social Security (IMSS)", "ISSSTE", "State-level ISSSTE", "Pemex", "Defensa",
  "Navy", "Seguro Popular", "IMSS PROSPERA", "Other institution"
)

# Tidy your model
tidy_model_meds <- tidy(model_meds, conf.int = TRUE)

# Filter for provider terms only
tidy_providers_meds <- tidy_model_meds %>%
  filter(term %in% provider_map$term) %>%
  left_join(provider_map, by = "term") %>%
  arrange(estimate)

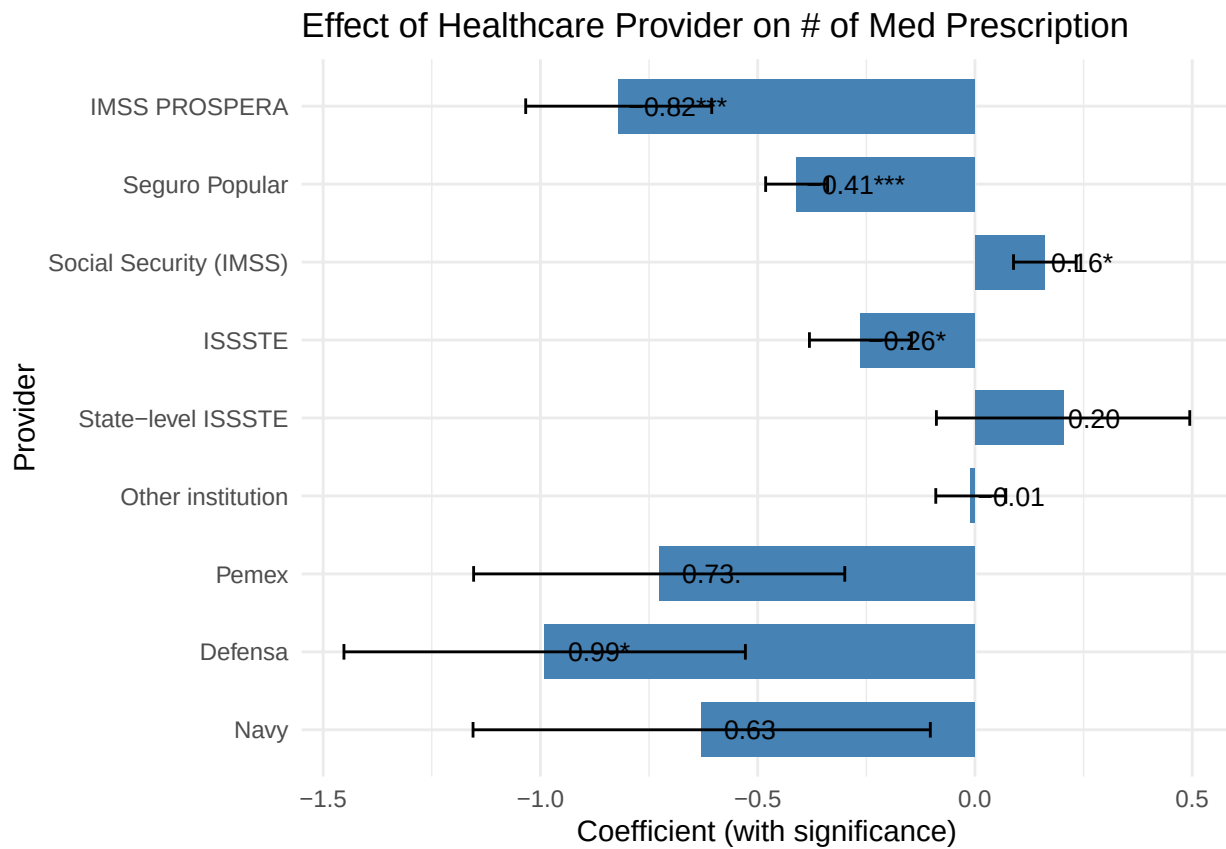
# Join the results on quality by provider
tidy_providers_meds <- tidy_providers_meds %>%
  left_join(provider_summary, by = "provider")

# 1. Add significance stars
tidy_providers_meds <- tidy_providers_meds %>%
  mutate(
    stars = case_when(
      p.value < 0.001 ~ "***",
      p.value < 0.01 ~ "**",
      p.value < 0.05 ~ "*",
      p.value < 0.1 ~ ".",
      TRUE ~ ""
    ),
    coef_label = paste0(sprintf("%.2f", estimate), stars)
  )

# 2. Order by average quality (ensure provider is factor)
tidy_providers_meds <- tidy_providers_meds %>%
  mutate(provider = fct_reorder(provider, avg_quality))

# 3. Plot
ggplot(tidy_providers_meds, aes(x = provider, y = estimate)) +
  geom_col(fill = "steelblue", width = 0.7) +
  geom_errorbar(aes(ymin = estimate - std.error,
                    ymax = estimate + std.error), width = 0.2) +
  geom_text(aes(label = coef_label), hjust = -0.1, size = 3.5) +
  coord_flip() +
  labs(
    title = "Effect of Healthcare Provider on # of Med Prescription",
    x = "Provider",
    y = "Coefficient (with significance)"
  ) +
  theme_minimal()

```



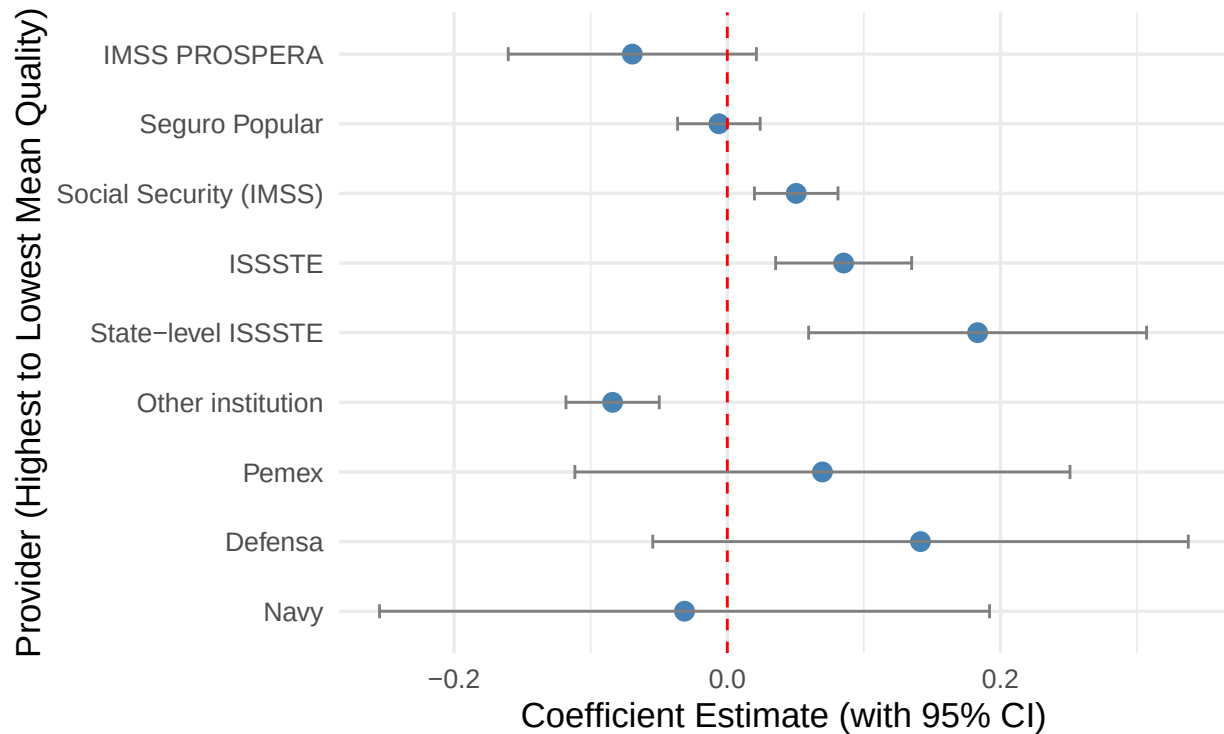
```
plot_model(model_tests, show.values = TRUE, value.offset = 0.3) +
  coord_flip()
```

```
## Coordinate system already present. Adding new coordinate system, which will
## replace the existing one.
```



```
ggplot(tidy_providers, aes(x = estimate, y = provider)) +
  geom_point(color = "steelblue", size = 3) +
  geom_errorbarh(aes(xmin = conf.low, xmax = conf.high), height = 0.2, color = "gray50") +
  geom_vline(xintercept = 0, linetype = "dashed", color = "red") +
  labs(
    title = "Effect of Provider Type on Number of Med Prescription",
    subtitle = "Ordered by Mean Quality of Care",
    x = "Coefficient Estimate (with 95% CI)",
    y = "Provider (Highest to Lowest Mean Quality)"
  ) +
  theme_minimal(base_size = 13)
```

Effect of Provider Type on Number of Med Presc Ordered by Mean Quality of Care



Hypo 2: Low effort per treatment

Without capitation or fee-for-service incentives and with low monitoring, providers put less effort per patient.

```
health_quality_data <- health_quality_data %>%
  # Consultation time in minutes
  mutate(consult_time = as.numeric(P4_7H) * 60 + as.numeric(P4_7M),

  # Effort perception variables (reverse coded so higher = more effort)
  ask_questions = ifelse(P8_5_2 %in% 1:4, 5 - as.numeric(P8_5_2), NA),
  enough_time = ifelse(P8_5_3 %in% 1:4, 5 - as.numeric(P8_5_3), NA),
  shared_decision = ifelse(P8_5_4 %in% 1:4, 5 - as.numeric(P8_5_4), NA),
  explain_clear = ifelse(P8_5_5 %in% 1:4, 5 - as.numeric(P8_5_5), NA),
  coordinate_specialist = ifelse(P8_5_6 %in% 1:4, 5 - as.numeric(P8_5_6), NA),

  # Other effort-relevant measures
  review_meds = ifelse(P8_8_1 == 1, 1, ifelse(P8_8_1 == 2, 0, NA)),
  explain_sidefx = ifelse(P8_8_2 == 1, 1, ifelse(P8_8_2 == 2, 0, NA)),

  # Provider recoding (if not yet done)
  provider = factor(P3_7, levels = c(1,2,3,4,5,6,7,8,9,10,11),
    labels = c("IMSS", "ISSSTE", "State-level ISSSTE", "Pemex", "Defensa", "Navy",
      "IMSS PROSPERA", "Other", "Private", "Other")),

  sex = factor(SEX0, labels = c("Male", "Female")),
  age = as.numeric(EDAD))
```

```

)

# Standardize variables
effort_vars <- c("consult_time", "ask_questions", "enough_time", "shared_decision",
                "explain_clear", "coordinate_specialist", "review_meds", "explain_sidefx")

health_quality_data <- health_quality_data %>%
  mutate(across(all_of(effort_vars), ~ scale(.)[,1], .names = "z_{col}")) %>%
  rowwise() %>%
  mutate(effort_index = mean(c_across(starts_with("z_")), na.rm = TRUE)) %>%
  ungroup()

# Define the base comparison for the Private
health_quality_data$provider <- relevel(health_quality_data$provider, ref = "Private")

# Define the model
effort_model <- lm(effort_index ~ provider + age + sex, data = health_quality_data)
effort_model_conditions <- lm(effort_index ~ provider + age + sex + condition, data = health_quality_data)

# Print the summary
summary(effort_model)

```

```

##
## Call:
## lm(formula = effort_index ~ provider + age + sex, data = health_quality_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.9318 -0.4156  0.2383  0.5803 11.5827
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    0.1387150   0.0247591    5.603 2.19e-08 ***
## providerIMSS   -0.2047115   0.0255570   -8.010 1.34e-15 ***
## providerISSSTE -0.1637771   0.0423519   -3.867 0.000111 ***
## providerState-level ISSSTE -0.2297185   0.1057637   -2.172 0.029890 *
## providerPemex    0.1374223   0.1546050    0.889 0.374109
## providerDefensa  0.2376824   0.1655521    1.436 0.151135
## providerNavy     0.3817271   0.1908562    2.000 0.045531 *
## providerSeguro Popular -0.2239564   0.0253566   -8.832 < 2e-16 ***
## providerIMSS PROSPERA -0.2899166   0.0772836   -3.751 0.000177 ***
## providerOther    -0.2102375   0.0286598   -7.336 2.46e-13 ***
## age              0.0006501   0.0003765    1.727 0.084261 .
## sexFemale        0.0152177   0.0184813    0.823 0.410301
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.7358 on 6832 degrees of freedom
## (721 observations deleted due to missingness)
## Multiple R-squared:  0.01803,    Adjusted R-squared:  0.01645
## F-statistic: 11.41 on 11 and 6832 DF,  p-value: < 2.2e-16

```

```
summary(effort_model_conditions)
```

```
##
## Call:
## lm(formula = effort_index ~ provider + age + sex + condition,
##     data = health_quality_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.0128 -0.4044  0.2374  0.5316 11.6111
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    0.1668521   0.0344452   4.844 1.30e-06 ***
## providerIMSS   -0.2317853   0.0261501  -8.864 < 2e-16 ***
## providerISSSTE -0.1857848   0.0425661  -4.365 1.29e-05 ***
## providerState-level ISSSTE -0.2450798   0.1057658  -2.317 0.02052 *
## providerPemex   0.1631218   0.1549331   1.053 0.29245
## providerDefensa  0.2510766   0.1675834   1.498 0.13412
## providerNavy    0.3666575   0.1908962   1.921 0.05481 .
## providerSeguro Popular -0.2464204   0.0258368  -9.538 < 2e-16 ***
## providerIMSS PROSPERA -0.3172344   0.0776605  -4.085 4.46e-05 ***
## providerOther   -0.2219206   0.0292056  -7.599 3.40e-14 ***
## age             0.0003838   0.0004276   0.898 0.36944
## sexFemale       0.0042990   0.0188258   0.228 0.81938
## condition2      0.3431007   0.1419054   2.418 0.01564 *
## condition3     -0.0129738   0.1089130  -0.119 0.90518
## condition4      0.0281376   0.0376349   0.748 0.45470
## condition5     -0.4734227   0.5267361  -0.899 0.36880
## condition6     -0.1224395   0.1075980  -1.138 0.25519
## condition7      0.0382414   0.1416341   0.270 0.78717
## condition8      0.0041364   0.1037949   0.040 0.96821
## condition9      0.0886441   0.0739141   1.199 0.23046
## condition10     0.0534428   0.0912273   0.586 0.55802
## condition11     0.0420197   0.1710026   0.246 0.80590
## condition12     0.0824640   0.0441958   1.866 0.06210 .
## condition13     0.0946965   0.0464650   2.038 0.04159 *
## condition14    -0.3280695   0.2240909  -1.464 0.14324
## condition15    -0.3804845   0.2465349  -1.543 0.12280
## condition16     0.0071723   0.0995280   0.072 0.94255
## condition17    -0.0061269   0.0583253  -0.105 0.91634
## condition18    -0.0365448   0.0682872  -0.535 0.59255
## condition19    -0.0532238   0.0949231  -0.561 0.57502
## condition20    -0.1393588   0.1223302  -1.139 0.25466
## condition21    -0.2346543   0.3296427  -0.712 0.47659
## condition22     0.0016008   0.0723973   0.022 0.98236
## condition23    -0.0638484   0.0716662  -0.891 0.37301
## condition24     0.1625321   0.2463818   0.660 0.50948
## condition25     0.0628270   0.2342656   0.268 0.78856
## condition26     0.6404500   0.7351777   0.871 0.38370
## condition28     0.0719035   0.1708361   0.421 0.67385
## condition29    -0.1830934   0.2466048  -0.742 0.45784
## condition30    -0.2595638   0.7350897  -0.353 0.72402
```

```
## condition32          0.3336212  0.5203584  0.641  0.52146
## condition33          -0.0134959  0.0500813 -0.269  0.78757
## condition34          -0.1263608  0.1754816 -0.720  0.47150
## condition36          -0.0252212  0.1258296 -0.200  0.84114
## condition37           0.1645347  0.1500384  1.097  0.27285
## condition38          -0.0327522  0.0997448 -0.328  0.74265
## condition39          -0.1949171  0.0683920 -2.850  0.00439 **
## condition40          -0.1386605  0.0883905 -1.569  0.11676
## condition41          -0.0817820  0.1211321 -0.675  0.49961
## condition42          -0.0567392  0.3014786 -0.188  0.85072
## condition43           0.1160555  0.0606573  1.913  0.05575 .
## condition44           0.1197687  0.0777759  1.540  0.12363
## condition45          -0.0279936  0.0335035 -0.836  0.40344
## condition99          -0.7295308  0.2345286 -3.111  0.00187 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.7343 on 6790 degrees of freedom
## (721 observations deleted due to missingness)
## Multiple R-squared:  0.02814,    Adjusted R-squared:  0.02056
## F-statistic:  3.71 on 53 and 6790 DF,  p-value: < 2.2e-16
```

```
## -- FOR PROVIDERS --
```

```
pred_df <- health_quality_data %>%
  group_by(provider) %>%
  summarise(mean_effort = mean(effort_index, na.rm = TRUE),
            se = sd(effort_index, na.rm = TRUE)/sqrt(n()))

# First get the original plot with sorting
plot <- plot_model(effort_model,
  type = "est",
  sort.est = TRUE, # Sort by effect size
  title = "Effects on Provider Effort Index",
  show.values = TRUE,
  show.p = TRUE,
  value.offset = 0.3,
  colors = c("firebrick", "steelblue")) # Red for negative, blue for positive

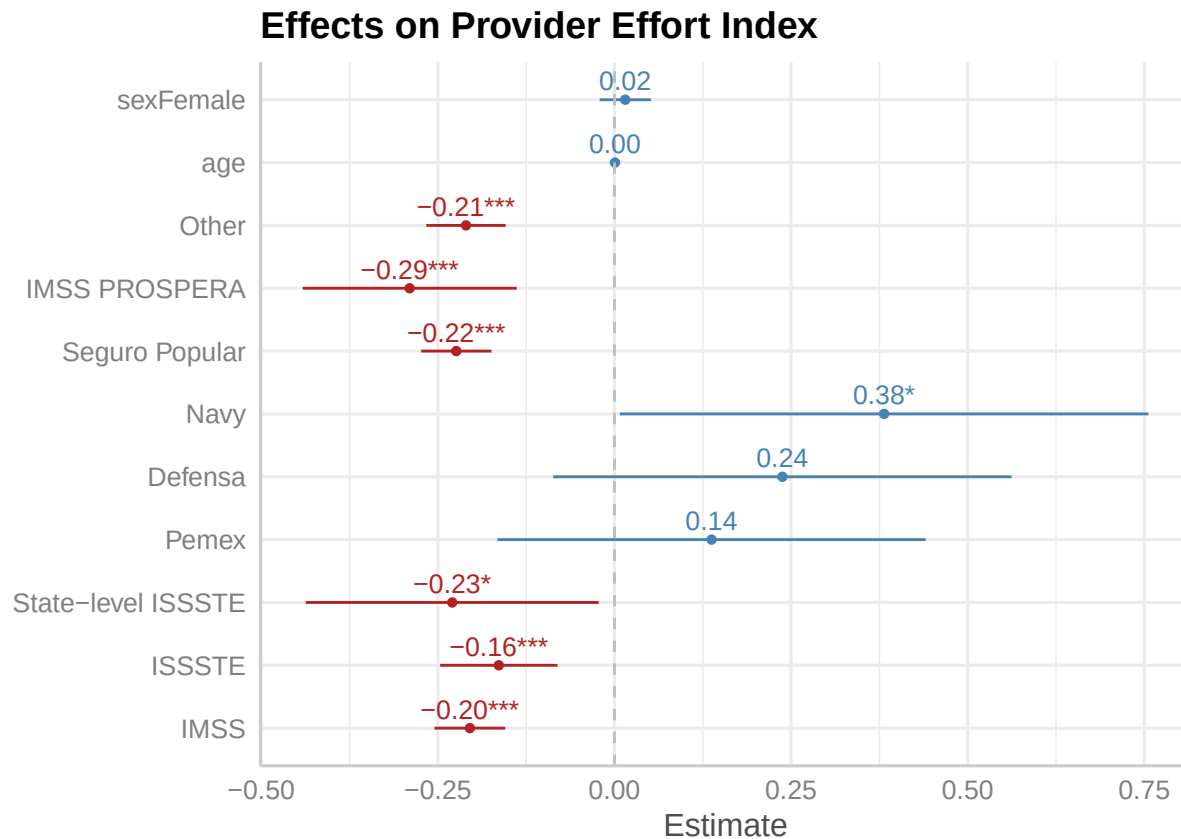
# Extract and modify the data while preserving order
plot_data <- plot$data
# Store original order
plot_data$original_order <- 1:nrow(plot_data)
# Remove "provider" from labels
plot_data$term <- gsub("provider", "", plot_data$term)

# Rebuild plot with modified labels but preserved order
ggplot(plot_data, aes(x = reorder(term, original_order), y = estimate, color = estimate > 0)) +
  geom_pointrange(aes(ymin = conf.low, ymax = conf.high), fatten = 1) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "gray") +
  scale_color_manual(values = c("firebrick", "steelblue")) +
  geom_text(aes(label = paste0(sprintf("%.2f", estimate),
                                ifelse(p.value < 0.001, "***",
                                ifelse(p.value < 0.01, "**",
```

```

                                ifelse(p.value < 0.05, "*", ""))))),
      vjust = -0.5, size = 3.5) +
labs(title = "Effects on Provider Effort Index",
      y = "Estimate",
      x = "") +
theme_sjplot() +
theme(axis.text.y = element_text(size = 10),
      plot.title = element_text(face = "bold", size = 14),
      legend.position = "none") +
coord_flip()

```



```

# -- FOR HEALTH CONDITIONS --
condition_map <- data.frame(
  term = paste0("condition", c(
    2, 3, 4, 5, 6, 7, 8, 9, 10, 11,
    12, 13, 14, 15, 16, 17, 18, 19, 20, 21,
    22, 23, 24, 25, 26, 28, 29, 30, 32, 33,
    34, 36, 37, 38, 39, 40, 41, 42, 43, 44,
    45, 99)
),
  condition = c(
    "Pneumonia", "COPD", "Cold/Sore throat", "Tuberculosis", "Ear infection", "Conjunctivitis", "Asthma",
    "Rheumatic fever", "Diabetes", "Hypertension", "Stroke", "Obesity", "Arthritis", "Diarrhea", "Gastr",
    "Hepatitis", "Kidney disease", "UTI", "Exanthematous disease", "STI", "HIV/AIDS", "Dengue", "Poison",
    "Accidental injury", "Injury from aggression", "Stress", "Depression", "Skin issues", "Oral disease",
    "Folk illnesses", "Pregnancy", "Cancer", "Other", "Don't know"
  )
)

```

```

)
)

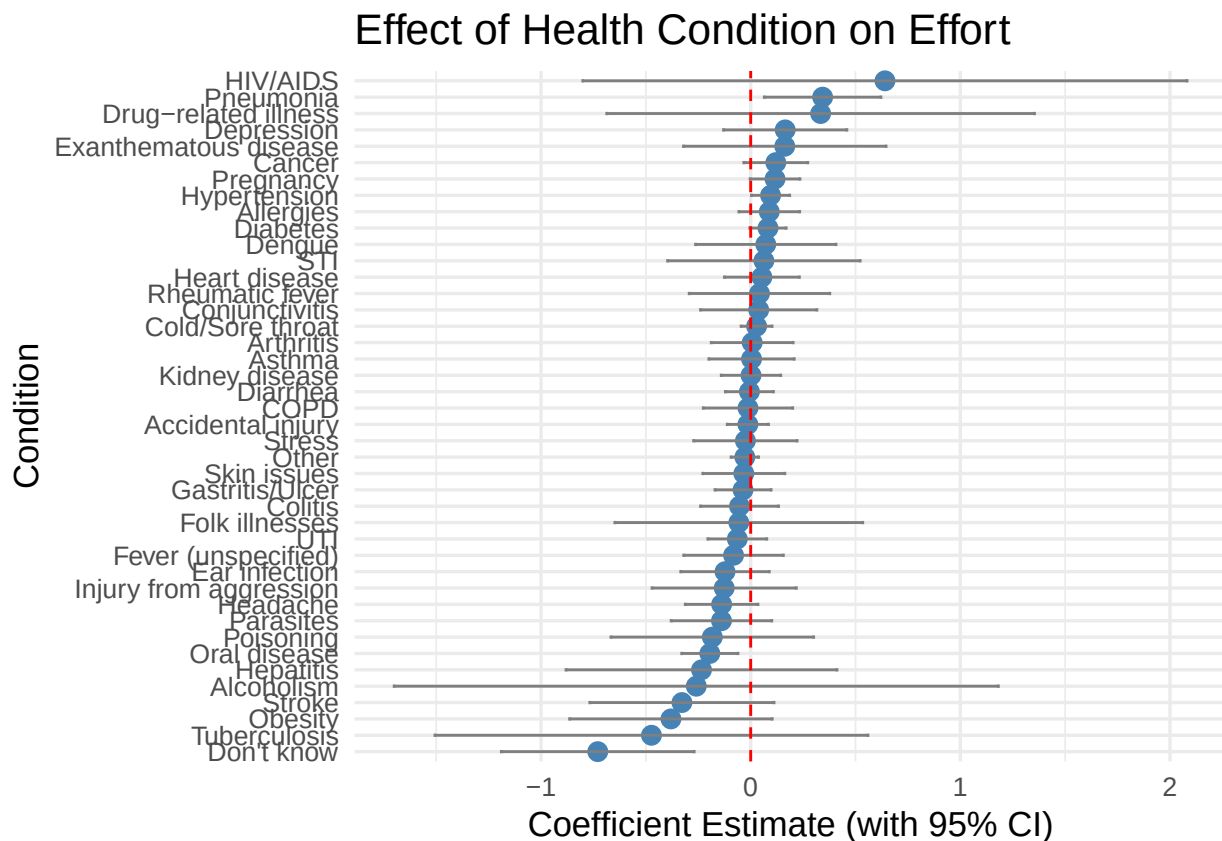
tidy_model_effort <- tidy(effort_model_conditions, conf.int = TRUE)

# Filter only condition variables
tidy_model_effort <- tidy_model_effort %>%
  filter(grepl("^condition\\d+", term)) %>%
  left_join(condition_map, by = "term") %>%
  arrange(estimate)

# Create factor with ordering for plotting
tidy_model_effort$condition <- factor(tidy_model_effort$condition, levels = tidy_model_effort$condition)

ggplot(tidy_model_effort, aes(x = estimate, y = condition)) +
  geom_point(color = "steelblue", size = 3) +
  geom_errorbarh(aes(xmin = conf.low, xmax = conf.high), height = 0.2, color = "gray50") +
  geom_vline(xintercept = 0, linetype = "dashed", color = "red") +
  labs(
    title = "Effect of Health Condition on Effort",
    x = "Coefficient Estimate (with 95% CI)",
    y = "Condition"
  ) +
  theme_minimal(base_size = 13)

```



Hypo 3: Moral hazard leads to inefficient care, not better care.

```
## ----- HYPOTHESIS 3 -----  
## Moral hazard leads to inefficient care, not better care.  
  
# Outcome: would return (Yes = 1)  
health_quality_data <- health_quality_data %>%  
  mutate(  
    health_outcome = ifelse(P7_1 %in% c(1,2), 1, 0)  
  )  
  
health_quality_data <- health_quality_data %>%  
  mutate(  
    num_meds = P5_1  
  )  
  
health_quality_data <- health_quality_data %>%  
  filter(!is.na(provider))    # remove any missing-provider rows  
  
# 1) Ensure provider is coded and labeled  
health_quality_data <- health_quality_data %>%  
  mutate(  
    provider = factor(provider, levels = unique(provider)), # preserve existing order  
    system    = recode(as.character(provider),  
      "1" = "Social Security (IMSS)",  
      "2" = "ISSSTE",  
      "3" = "State-level ISSSTE",  
      "4" = "Pemex",  
      "5" = "Defensa",  
      "6" = "Navy",  
      "7" = "Seguro Popular",  
      "8" = "IMSS PROSPERA",  
      "9" = "Private insurance",  
      "10" = "Other institution",  
      "11" = "Not affiliated",  
      "99" = "Doesn't know")  
  )  
  
# 2) TEST_REQUESTED: run reg within each provider, pull slope + CI  
test_df <- health_quality_data %>%  
  group_by(provider, system) %>%  
  do(tidy(  
    lm(health_outcome ~ test_requested + age + sex + factor(primary_condition), data = .),  
    conf.int = TRUE  
  )) %>%  
  filter(term == "test_requested") %>%  
  mutate(original_order = row_number())  
  
# 3) NUM_MEDS: same but for num_meds  
meds_df <- health_quality_data %>%  
  group_by(provider, system) %>%  
  do(tidy(  
    lm(health_outcome ~ num_meds + age + sex + factor(primary_condition), data = .),
```

```

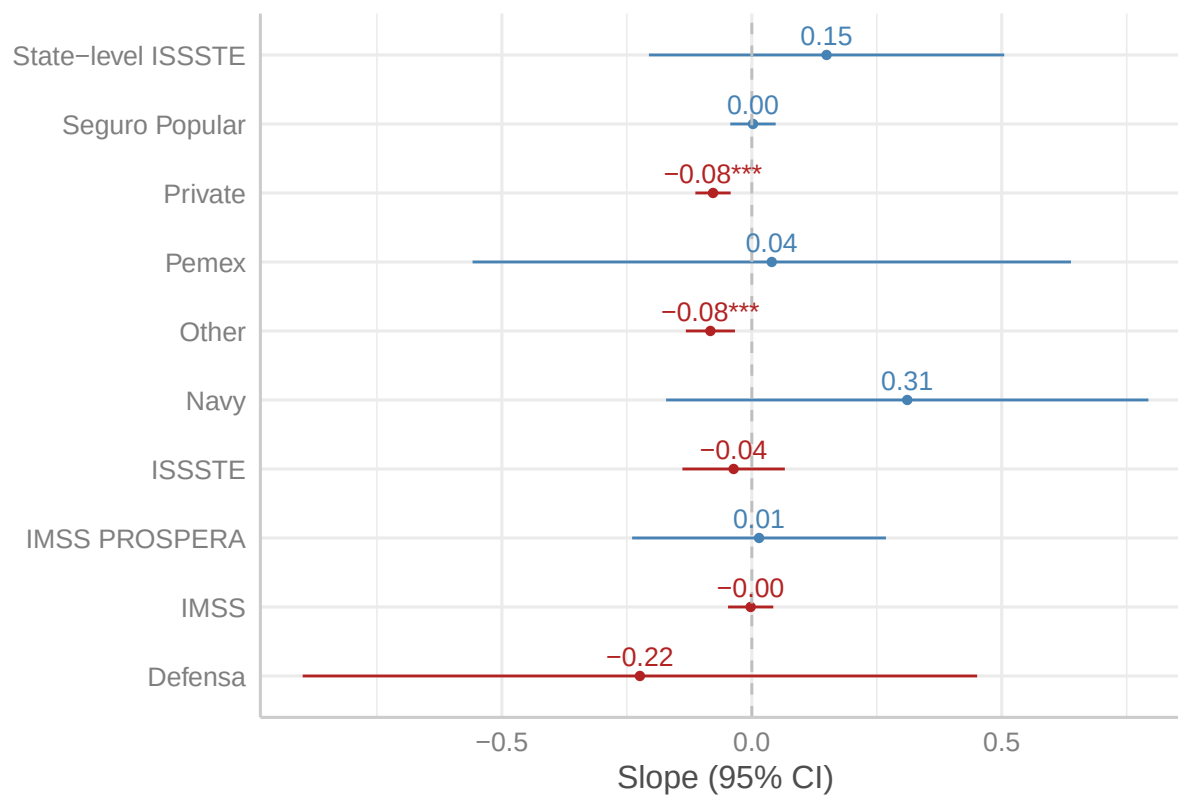
    conf.int = TRUE
  )) %>%
  filter(term == "num_meds") %>%
  mutate(original_order = row_number())

# 4) Function to plot any of these
plot_by_system <- function(df, title_text) {
  ggplot(df,
    aes(x = reorder(system, original_order),
        y = estimate,
        color = estimate > 0)) +
  geom_pointrange(aes(ymin = conf.low, ymax = conf.high),
    fatten = 1) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "gray") +
  scale_color_manual(values = c("firebrick", "steelblue")) +
  geom_text(aes(label = paste0(
    sprintf("%.2f", estimate),
    ifelse(p.value < 0.001, "****",
    ifelse(p.value < 0.01, "***",
    ifelse(p.value < 0.05, "*", ""))
  )
  ),
    vjust = -0.5, size = 3.5) +
  labs(title = title_text,
    y = "Slope (95% CI)",
    x = "") +
  theme_sjplot() +
  theme(axis.text.y = element_text(size = 10),
    plot.title = element_text(face = "bold", size = 14),
    legend.position = "none") +
  coord_flip()
}

# 5) Draw them
plot_by_system(test_df, "Effect of Test Requested on Better Care, by Provider")

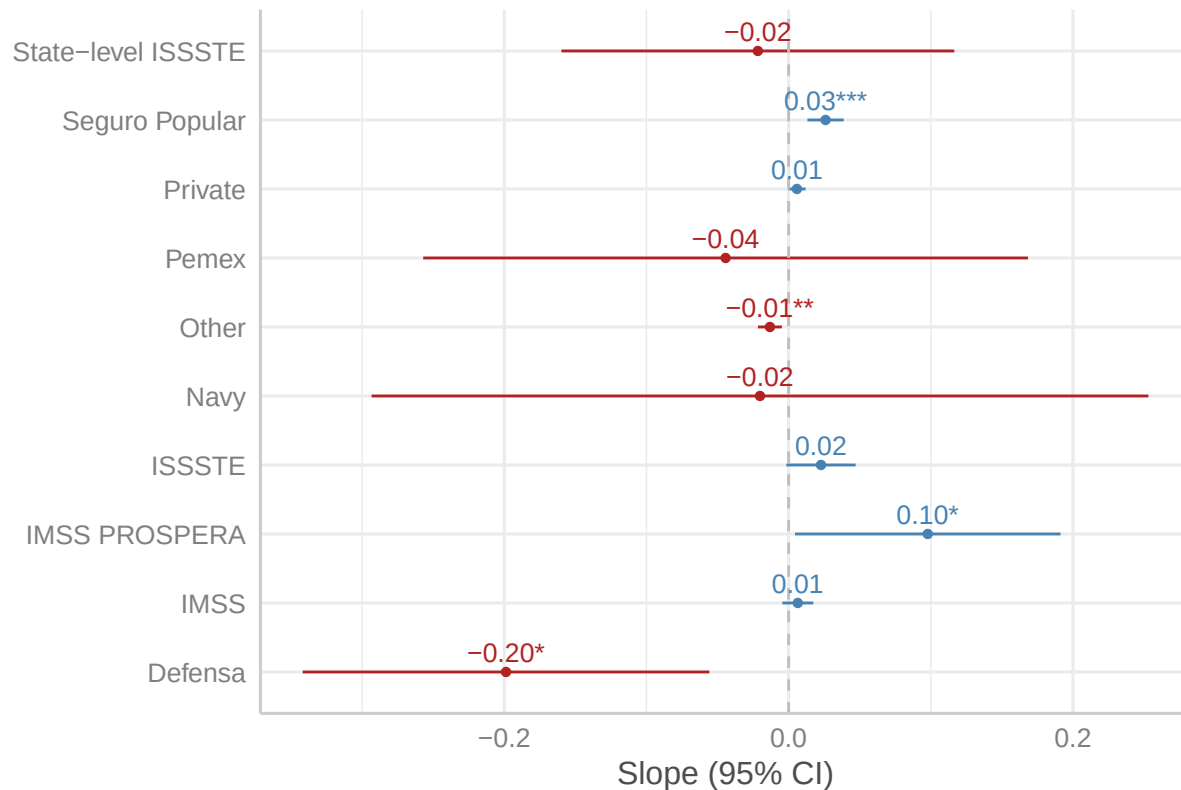
```

Effect of Test Requested on Better Care, by Provider



```
plot_by_system(meds_df, "Effect of Number of Medications on Better Care, by Provider")
```

Effect of Number of Medications on Better Care, by Pí



Does more test or more medication significantly improve health outcome?

```
# Outcome: would return (Yes = 1)
health_quality_data <- health_quality_data %>%
  mutate(
    health_outcome = ifelse(P7_2 %in% c(1), 1, 0)
  )

health_quality_data <- health_quality_data %>%
  mutate(
    num_meds = P5_1
  )

health_quality_data <- health_quality_data %>%
  filter(!is.na(provider)) # remove any missing-provider rows

# 1) Ensure provider is coded and labeled
health_quality_data <- health_quality_data %>%
  mutate(
    provider = factor(provider, levels = unique(provider)), # preserve existing order
    system = recode(as.character(provider),
      "1" = "Social Security (IMSS)",
      "2" = "ISSSTE",
      "3" = "State-level ISSSTE",
      "4" = "Pemex",
      "5" = "Defensa",
      "6" = "Navy",
```

```

        "7" = "Seguro Popular",
        "8" = "IMSS PROSPERA",
        "9" = "Private insurance",
        "10" = "Other institution",
        "11" = "Not affiliated",
        "99" = "Doesn't know")
    )

# 2) TEST_REQUESTED: run reg within each provider, pull slope + CI
test_df <- health_quality_data %>%
  group_by(provider, system) %>%
  do(tidy(
    lm(health_outcome ~ test_requested + age + sex + factor(primary_condition), data = .),
    conf.int = TRUE
  )) %>%
  filter(term == "test_requested") %>%
  mutate(original_order = row_number())

# 3) NUM_MEDS: same but for num_meds
meds_df <- health_quality_data %>%
  group_by(provider, system) %>%
  do(tidy(
    lm(health_outcome ~ num_meds + age + sex + factor(primary_condition), data = .),
    conf.int = TRUE
  )) %>%
  filter(term == "num_meds") %>%
  mutate(original_order = row_number())

# 4) Function to plot any of these
plot_by_system <- function(df, title_text) {
  ggplot(df,
    aes(x = reorder(system, original_order),
        y = estimate,
        color = estimate > 0)) +
  geom_pointrange(aes(ymin = conf.low, ymax = conf.high),
    fatten = 1) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "gray") +
  scale_color_manual(values = c("firebrick", "steelblue")) +
  geom_text(aes(label = paste0(
    sprintf("%.2f", estimate),
    ifelse(p.value < 0.001, "****",
    ifelse(p.value < 0.01, "***",
    ifelse(p.value < 0.05, "*", ""))
  )
  ),
    vjust = -0.5, size = 3.5) +
  labs(title = title_text,
    y = "Slope (95% CI)",
    x = "") +
  theme_sjplot() +
  theme(axis.text.y = element_text(size = 10),
    plot.title = element_text(face = "bold", size = 14),

```

```

    legend.position = "none") +
    coord_flip()
}

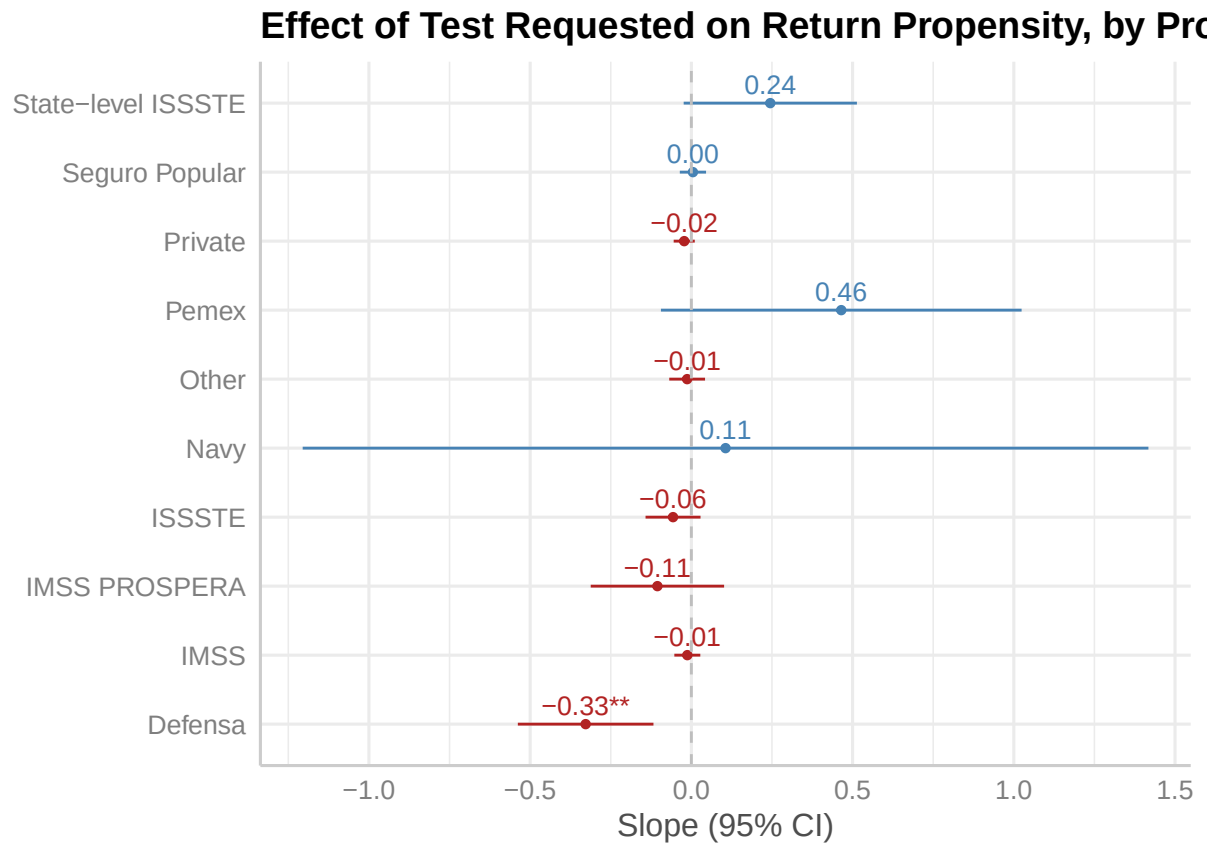
```

5) Draw them

```

plot_by_system(test_df, "Effect of Test Requested on Return Propensity, by Provider")

```

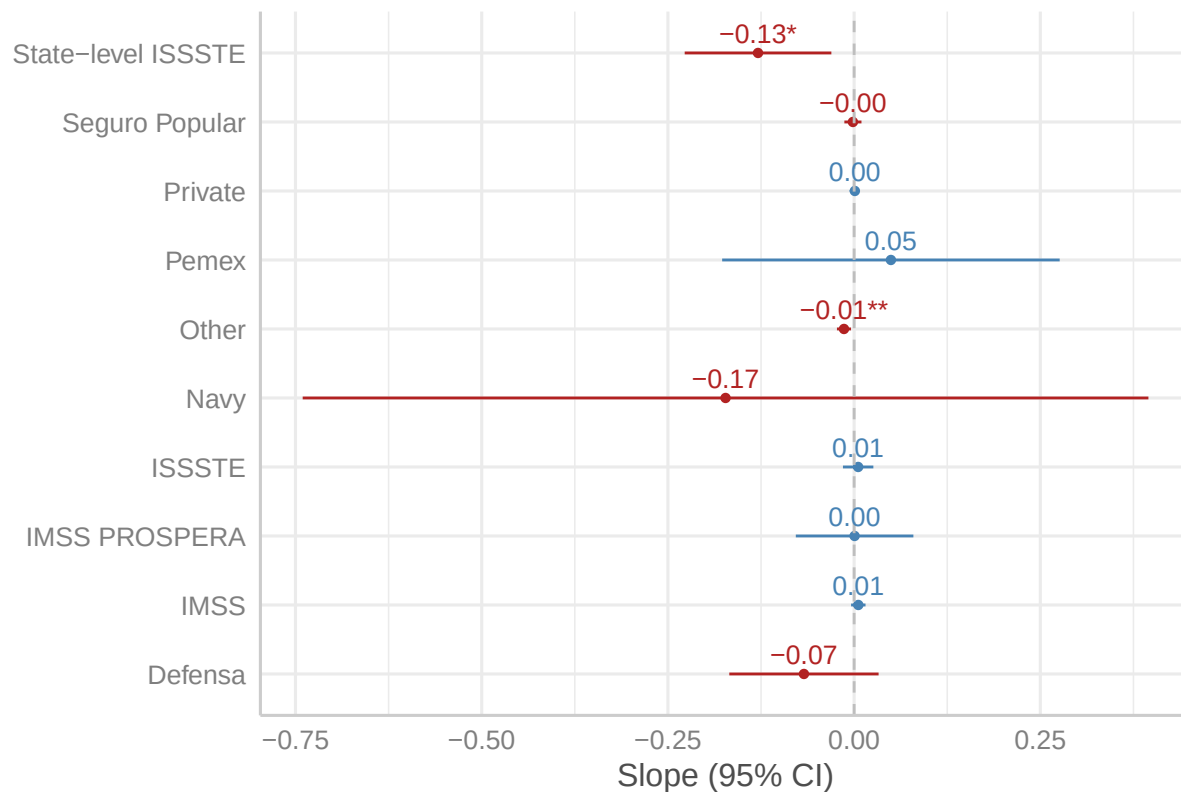


```

plot_by_system(meds_df, "Effect of Number of Medications on Return Propensity, by Provider")

```

Effect of Number of Medications on Return Propensity



Testing and patient out-of-pocket spending

```
# Total medical spending
health_quality_data <- health_quality_data %>%
  mutate(
    lab_cost = as.numeric(P6_4),
    med_cost = as.numeric(P5_6),
    other_cost = as.numeric(P6_6_1) + as.numeric(P6_6_2) + as.numeric(P6_6_3),
    total_cost = rowSums(across(c(lab_cost, med_cost, other_cost)), na.rm = TRUE)
  )

# Regression: Costs ~ Treatment Intensity

# 2) Extract test_requested slopes by provider
test_df <- health_quality_data %>%
  group_by(provider, system) %>%
  do(tidy(
    lm(total_cost ~ test_requested + age + sex + factor(primary_condition), data = .),
    conf.int = TRUE
  )) %>%
  filter(term == "test_requested") %>%
  mutate(original_order = row_number())

# 3) Extract num_meds slopes by provider
meds_df <- health_quality_data %>%
  group_by(provider, system) %>%
  do(tidy(
```

```

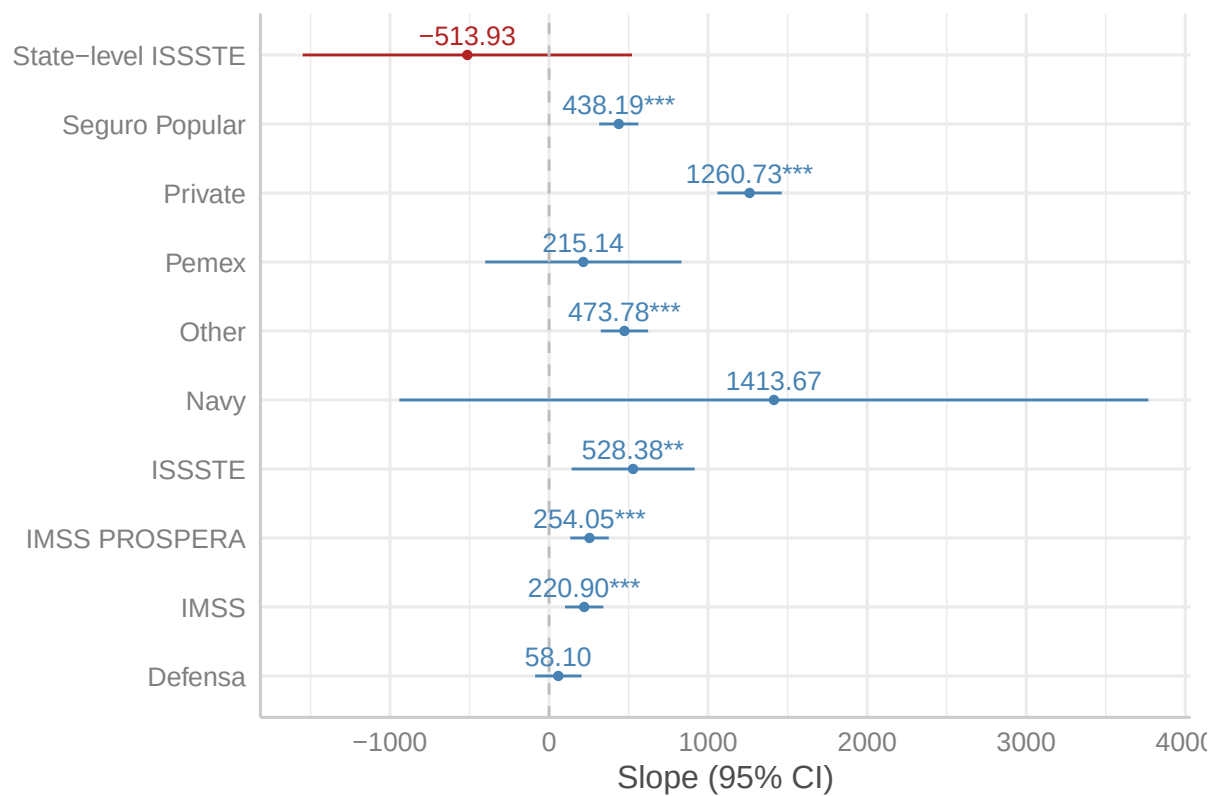
    lm(total_cost ~ num_meds + age + sex + factor(primary_condition), data = .),
    conf.int = TRUE
  )) %>%
  filter(term == "num_meds") %>%
  mutate(original_order = row_number())

# 4) Plotting function
plot_by_system <- function(df, title_text) {
  ggplot(df,
    aes(x = reorder(system, original_order),
        y = estimate,
        color = estimate > 0)) +
  geom_pointrange(aes(ymin = conf.low, ymax = conf.high),
    fatten = 1) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "gray") +
  scale_color_manual(values = c("firebrick", "steelblue")) +
  geom_text(aes(label = paste0(
    sprintf("%.2f", estimate),
    ifelse(p.value < 0.001, "***",
    ifelse(p.value < 0.01, "**",
    ifelse(p.value < 0.05, "*", ""))
  )
  ),
    vjust = -0.5, size = 3.5) +
  labs(title = title_text,
    y = "Slope (95% CI)",
    x = "") +
  theme_sjplot() +
  theme(axis.text.y = element_text(size = 10),
    plot.title = element_text(face = "bold", size = 14),
    legend.position = "none") +
  coord_flip()
}

# 5) Draw the two coefficient plots
plot_by_system(test_df, "Effect of Test Requested on Total Cost, by System")

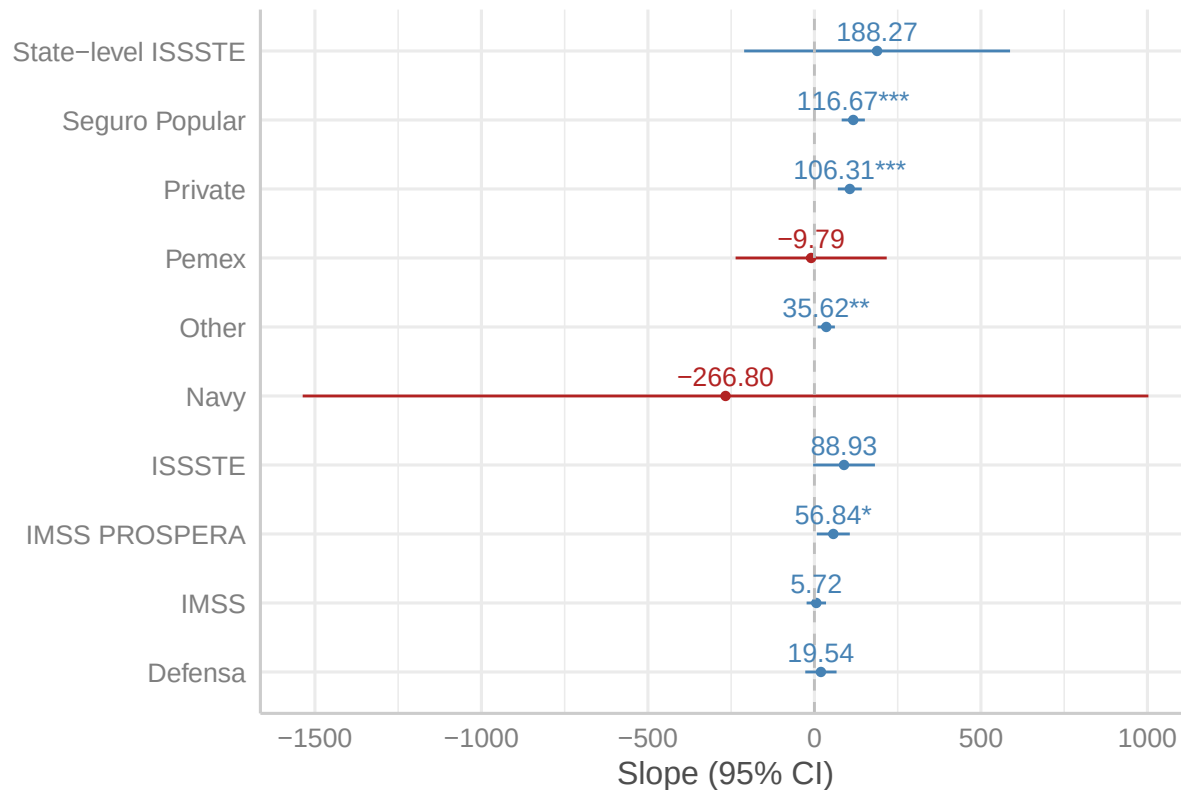
```

Effect of Test Requested on Total Cost, by System



```
plot_by_system(meds_df, "Effect of Number of Medications on Total Cost, by System")
```

Effect of Number of Medications on Total Cost, by Sys



Hypothesis 4

Wait time and access

```
health_quality_data <- health_quality_data %>%
  # convert factors/characters to numeric
  mutate(
    wait_time_rating = as.numeric(as.character(P4_6)),
    consult_minutes = as.numeric(P4_7H) * 60 + as.numeric(P4_7M),
    access_apr_rating = as.numeric(as.character(P8_2)),
    access_after_hours = as.numeric(as.character(P8_3)),
    access_by_phone = as.numeric(as.character(P8_4))
  )

# Quick check:
health_quality_data %>%
  select(wait_time_rating, consult_minutes,
         access_apr_rating, access_after_hours, access_by_phone) %>%
  summary()
```

```
## wait_time_rating consult_minutes access_apr_rating access_after_hours
## Min. :1.000 Min. : 1.00 Min. :1.000 Min. :1.000
## 1st Qu.:2.000 1st Qu.: 15.00 1st Qu.:1.000 1st Qu.:3.000
## Median :3.000 Median : 20.00 Median :1.000 Median :4.000
## Mean :3.083 Mean : 71.11 Mean :1.536 Mean :3.644
```

```
## 3rd Qu.:4.000      3rd Qu.: 30.00      3rd Qu.:2.000      3rd Qu.:5.000
## Max.      :9.000      Max.      :6039.00      Max.      :5.000      Max.      :5.000
##                                     NA's      :1125      NA's      :75
## access_by_phone
## Min.      :1.000
## 1st Qu.:3.000
## Median :4.000
## Mean      :3.863
## 3rd Qu.:5.000
## Max.      :5.000
## NA's      :66
```

Wait time and access - plot

```
# 1) Compute wait_time_score & consult_minutes
health_quality_data <- health_quality_data %>%
  mutate(
    wait_time_rating = as.numeric(as.character(P4_6)),
    wait_time_score  = 6 - wait_time_rating,           # higher = worse wait
    consult_minutes  = as.numeric(P4_7H) * 60 + as.numeric(P4_7M),
    provider         = factor(P3_7,
                              levels = c(9,1,2,3,4,5,6,7,8,10,11), # put 9="Private" first
                              labels = c(
                                "Private insurance",
                                "Social Security (IMSS)", "ISSSTE", "State-level ISSSTE",
                                "Pemex", "Defensa", "Navy", "Seguro Popular",
                                "IMSS PROSPERA", "Other institution", "Not affiliated"
                              )),
    primary_condition = factor(P1_2),
    sex               = factor(SEX0, labels = c("Male", "Female")),
    age               = as.numeric(EDAD)
  )

# 2) Model 1: Wait time
mod_wait <- lm(wait_time_score ~ provider + age + sex + factor(primary_condition),
               data = health_quality_data)

wait_coefs <- tidy(mod_wait, conf.int = TRUE) %>%
  filter(grepl("^provider", term)) %>%
  # drop the reference level (intercept) row if it sneaks in
  filter(term != "(Intercept)") %>%
  mutate(
    system      = sub("^provider", "", term),
    system      = gsub("\\.", " ", system), # clean up label if needed
    original_order = row_number()
  )

# 3) Model 2: Consultation minutes
mod_consult <- lm(consult_minutes ~ provider + age + sex + factor(primary_condition),
                 data = health_quality_data)

consult_coefs <- tidy(mod_consult, conf.int = TRUE) %>%
  filter(grepl("^provider", term)) %>%
  filter(term != "(Intercept)") %>%
```

```

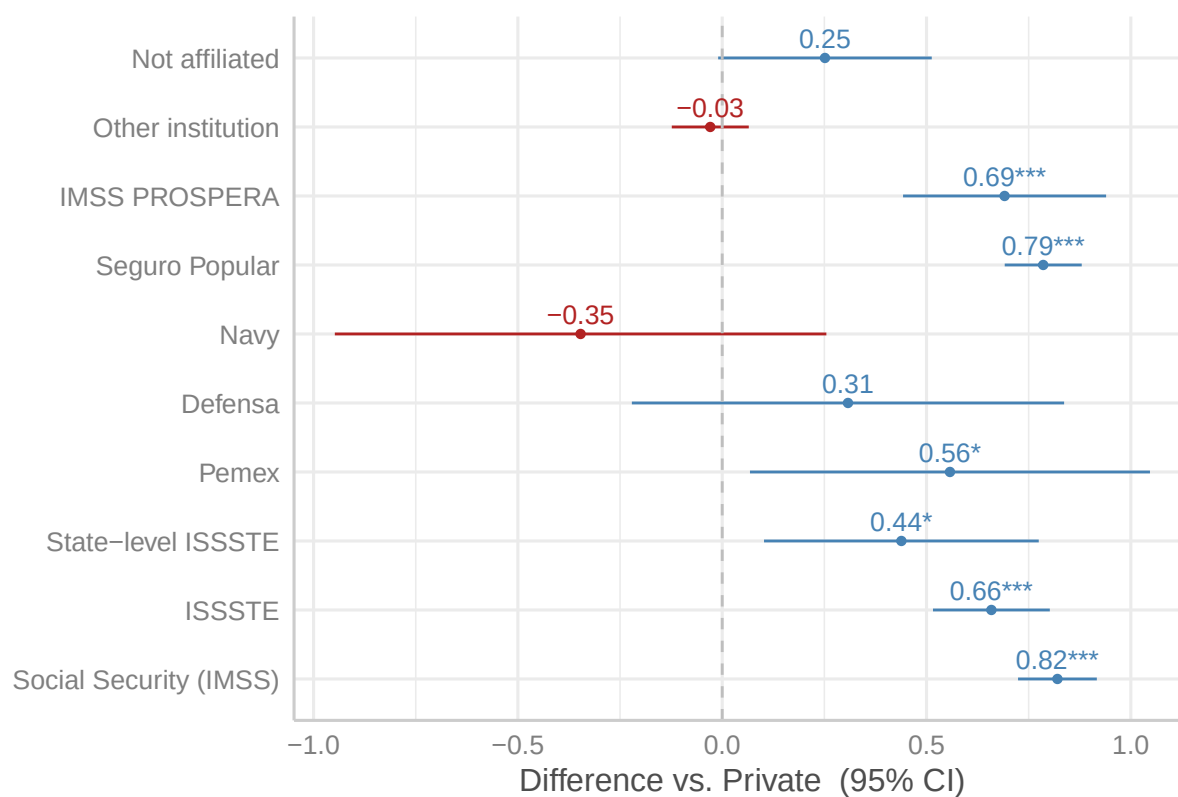
mutate(
  system      = sub("^provider", "", term),
  system      = gsub("\\.", " ", system),
  original_order = row_number()
)

# 4) Plotting function
plot_coef <- function(df, title_text) {
  ggplot(df, aes(x = reorder(system, original_order),
    y = estimate,
    color = estimate > 0)) +
  geom_pointrange(aes(ymin = conf.low, ymax = conf.high),
    fatten = 1) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "gray") +
  scale_color_manual(values = c("firebrick", "steelblue")) +
  geom_text(aes(label = paste0(
    sprintf("%.2f", estimate),
    ifelse(p.value < 0.001, "***",
    ifelse(p.value < 0.01, "**",
    ifelse(p.value < 0.05, "*", ""))
  )
  ),
    vjust = -0.5, size = 3.5) +
  labs(title = title_text,
    y = "Difference vs. Private (95% CI)",
    x = "") +
  theme_sjplot() +
  theme(axis.text.y = element_text(size = 10),
    plot.title = element_text(face = "bold", size = 14),
    legend.position = "none") +
  coord_flip()
}

# 5) Draw the two plots
plot_coef(wait_coefs, "Difference in Wait Time (versus Private)")

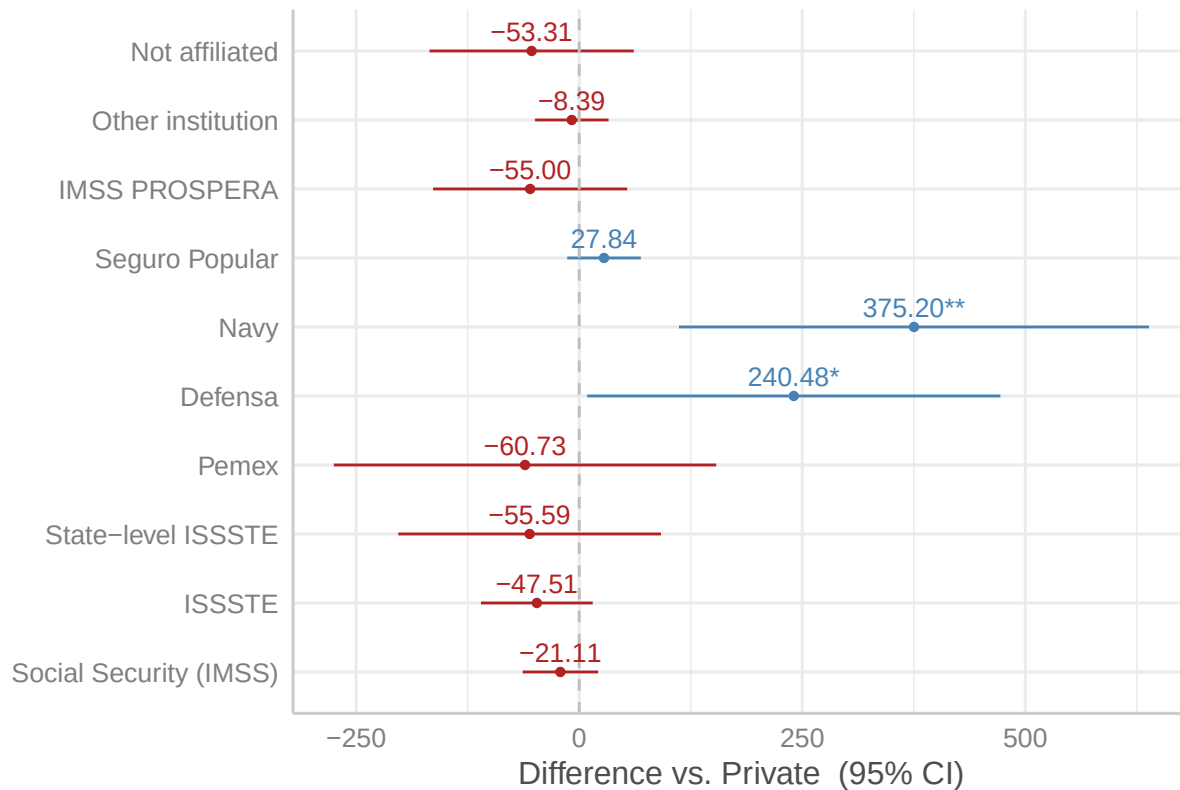
```

Difference in Wait Time (versus Private)



```
plot_coef(consult_coefs, "Difference in Consult Minutes (versus Private)")
```

Difference in Consult Minutes (versus Private)



Draft code.

```
health_quality_data$provider <- relevel(health_quality_data$provider, ref = "Private insurance")

provider_model <- lm(
  health_outcome ~ provider * effort_index + age + sex + primary_condition,
  data = health_quality_data
)

provider_df <- tidy(provider_model, conf.int = TRUE) %>%
  filter(grepl("^provider", term)) %>%
  # drop the reference level (intercept) row if it sneaks in
  filter(term != "(Intercept)") %>%
  mutate(
    system      = sub("^provider", "", term),
    system      = gsub("\\.", " ", system), # clean up label if needed
    original_order = row_number()
  )

plot_by_system <- function(df, title_text) {
  ggplot(df, aes(x = fct_reorder(system, original_order), y = estimate, color = estimate > 0)) +
    geom_pointrange(aes(ymin = conf.low, ymax = conf.high), fatten = 1) +
    geom_hline(yintercept = 0, linetype = "dashed", color = "gray") +
    scale_color_manual(values = c("firebrick", "steelblue")) +
    geom_text(aes(label = paste0(
      sprintf("%.2f", estimate),

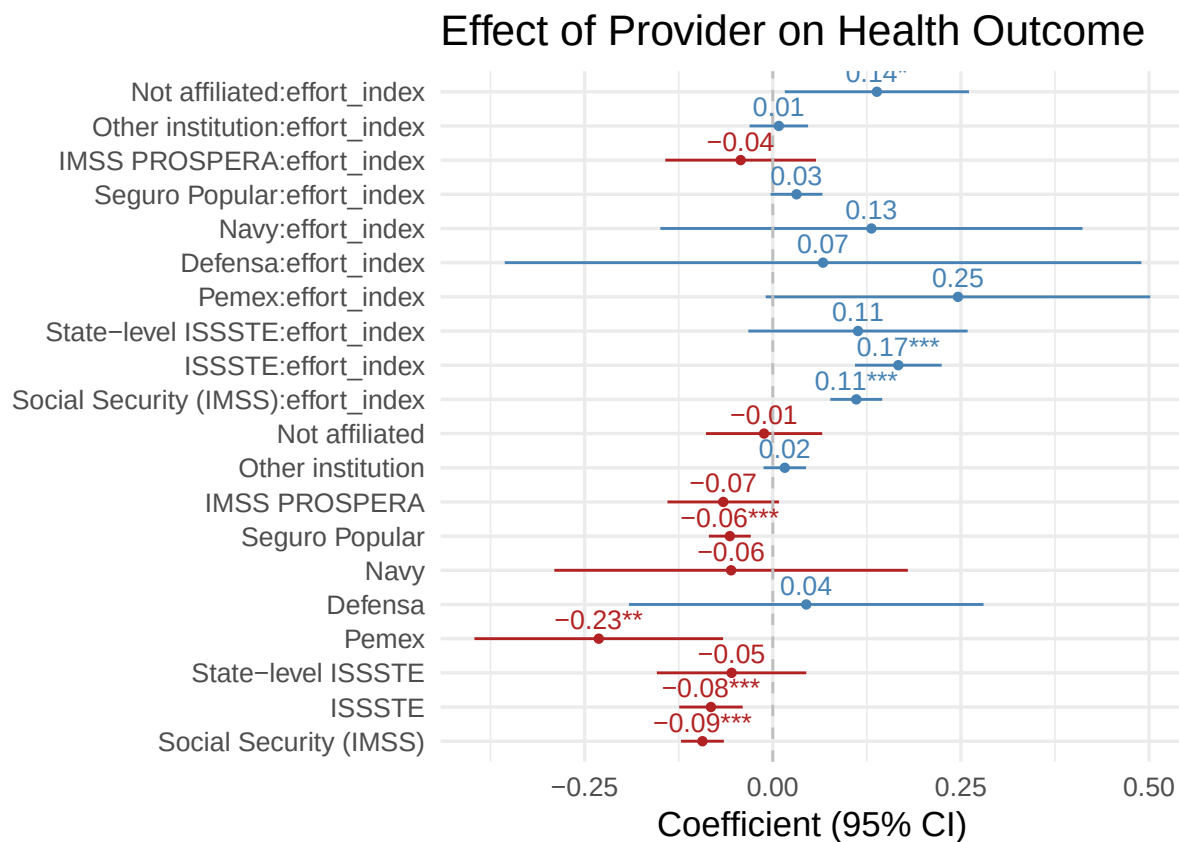
```

```

    ifelse(p.value < 0.001, "***",
           ifelse(p.value < 0.01, "**",
                  ifelse(p.value < 0.05, "*", ""))
    )
  )
  vjust = -0.5, size = 3.5) +
  labs(title = title_text, y = "Coefficient (95% CI)", x = "") +
  theme_minimal(base_size = 13) +
  theme(legend.position = "none") +
  coord_flip()
}

plot_by_system(provider_df, "Effect of Provider on Health Outcome")

```



Is our effort index robust? - Test whether effort index is positively and significantly contributing to health outcomes.

```

effort_df <- health_quality_data %>%
  group_by(provider, system) %>%
  do(tidy(
    lm(health_outcome ~ effort_index + age + sex + factor(primary_condition), data = .),
    conf.int = TRUE
  )) %>%
  filter(term == "effort_index") %>%
  mutate(original_order = row_number())

```

```

plot_by_system <- function(df, title_text) {
  ggplot(df,
    aes(x = reorder(system, original_order),
        y = estimate,
        color = estimate > 0)) +
  geom_pointrange(aes(ymin = conf.low, ymax = conf.high),
    fatten = 1) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "gray") +
  scale_color_manual(values = c("firebrick", "steelblue")) +
  geom_text(aes(label = paste0(
    sprintf("%.2f", estimate),
    ifelse(p.value < 0.001, "***",
    ifelse(p.value < 0.01, "**",
    ifelse(p.value < 0.05, "*", ""))
  )
  )),
    vjust = -0.5, size = 3.5) +
  labs(title = title_text,
    y = "Slope (95% CI)",
    x = "") +
  theme_sjplot() +
  theme(axis.text.y = element_text(size = 10),
    plot.title = element_text(face = "bold", size = 14),
    legend.position = "none") +
  coord_flip()
}

plot_by_system(effort_df, "Effect of Effort on Health Outcome, by System")

```

Effect of Effort on Health Outcome, by System

